A one-dimensional area law and a matrix product state approximation from a global spectral gap: the collar-free route

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We take the two September 2026 manuscripts *A two-dimensional area law from a global spectral gap* [1] and *Polynomial PEPS approximation of gapped square-grid ground states* [2] and run their arguments in one dimension. Two things happen. First, the area-law proof collapses: the marginal tail bound and the buffered conditional-entropy discount of [1, Section 2], together with the Araki–Lieb inequality, already prove $S(X) \le C(q,R)\,(1 + J/\Delta)$ for every interval X. The sub-volume recursion, the positive quasi-local constraints, the replica scanner, the amplification and the two-family geometry of [1] are not needed. The reason is geometric: a shell around an interval has a bounded number of crossing interactions, so the discount costs a constant instead of a boundary-sized error, and the buffer whose entropy must be controlled is a constant number of sites. Second, the tensor-network theorem survives as a linear *sweep* instead of a dyadic hierarchy. The encoded holes, adaptive patches, angular information estimates and row-column routing of [2] disappear; what remains are the Uhlmann-type birth and exchange moves, the information reset by a Gaussian filter, the dimension-free density compression and the column-extraction lemma. The result is a matrix product state of bond dimension CL^c with vector error L^{-1}. We also record that in one dimension the tail bound makes the matrix product state theorem a two-line corollary of the area law by sequential Schmidt truncation. Every statement in this note is written twice: once in words, once precisely. The area-law proof uses none of the techniques of Hastings or of Arad–Kitaev–Landau–Vazirani; the tensor-network proof uses a Lieb–Robinson bound in exactly one lemma, and we indicate how to avoid it at the price of a quasi-polynomial bond dimension.

I. OVERVIEW

In words. Both source papers are long because two dimensions is hard. The entanglement of a region is produced along its boundary, and in two dimensions the boundary of a region of linear size r has r sites: every estimate that costs "a boundary" costs r, and every buffer that one puts around a region to isolate it is itself a region whose entanglement must be controlled. The whole replica, scanner and amplification apparatus exists to break this circularity. In one dimension the boundary of an interval is two points. An estimate that costs a boundary costs a constant, and a buffer of constant width has, trivially, constant entropy. So the circularity never arises, and one short section of the two-dimensional paper proves the whole area law. This is the content of Part I.

The tensor-network theorem is a different story. It does not follow from entropy bounds alone (the papers explain why: an entropy bound on each cut does not give a global vector approximation). Its two-dimensional proof distributes the ground vector among "parties" attached to dyadic squares, by local changes of ownership whose errors are controlled by information estimates, then compresses the resulting distributed density into a tensor network. In one dimension a dyadic hierarchy is the wrong geometry: a binary tree drawn on a line has cutwidth $\log L$, which turns polynomial bond dimension into quasi-polynomial. The right geometry is a sweep from left to right with a moving "remainder" party. All the genuinely one-dimensional difficulties (what to do at the ends of a band when a third owner is adjacent) are solved by exactly the seven-move word trick that the two-dimensional paper uses along an edge. This is Part II. We also show that, in one dimension, the tail bound of Part I makes the tensor-network theorem a corollary of the area law by the classical sequential-truncation argument; the sweep protocol is the faithful one-dimensional analogue of the two-dimensional construction, and it is what generalizes.

Table I is a map from the two-dimensional proofs to this note. The reader who wants only the one-dimensional area law can read Sections II–V; this is about four pages.

Precisely. Throughout, $q \ge 2$ and $R \ge 1$ are integers and $J, \Delta > 0$. A finite set $\Lambda \subset \mathbb{Z}$ carries the induced path graph (edges between consecutive integers both in Λ) and graph distance d_Λ. Each site carries $\mathbb{C}^q$ and $\mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} \mathbb{C}^q$. An admissible Hamiltonian is

$$H = \sum_{\substack{\varnothing \ne X \subseteq \Lambda \\ \operatorname{diam}_{d_\Lambda} X \le R}} h_X, \qquad h_X = h_X^*, \qquad \|h_X\| \le J, \quad (1)$$

one summand per support, zero summands allowed. The standing spectral hypothesis is that H has a unique normalized ground vector Ω (up to phase), ground energy E_0, and

$$H - E_0 \mathrm{I} \;\ge\; \Delta\big(\mathrm{I} - |\Omega\rangle\langle\Omega|\big). \quad (2)$$

For $A \subseteq \Lambda$, $\rho_A = \operatorname{Tr}_{\Lambda\setminus A} |\Omega\rangle\langle\Omega|$, $S(A) = S_\Omega(A) = -\operatorname{Tr} \rho_A \log \rho_A$ with natural logarithms and $0 \log 0 = 0$, $S(A|B) = S(AB) - S(B)$ and $I(A:B) = S(A) + S(B) - S(AB)$ for disjoint A, B. The edge boundary is $\partial_\Lambda A = \{\{x, x+1\} \subseteq \Lambda : \text{ exactly one of } x, x+1 \text{ lies in } A\}$. Constants C, c depend only on q, R, J, Δ unless stated otherwise and may change from line to line; where a dependence is tracked it is said so. The two theorems of the note are the following.

Two-dimensional ingredient	Fate in one dimension
Marginal tail bound ([1], Lemma 2.1)	Kept verbatim; the cut parameter $\mathcal{B}$ is a constant (Section III).
Buffered conditional-entropy discount ([1], Lemma 2.2)	Kept; the buffer has *constant* width and the error is a constant (Section IV).
Sub-volume recursion, exponent $e_0 < 1$	Unnecessary: Araki–Lieb closes the argument directly (Section V). Sketched as a "long route" in Section VI.
Positive quasi-local constraints, truncation	Unnecessary for the area law; not used for the MPS either (replaced by the Gaussian reset).
Conditional estimates, replicas, transport, comparators, scanner	Unnecessary.
Amplification by random constraint products	Unnecessary; Section VI explains how it would be used.
Two families of regions	Trivial on a line; Section VI.
Coarse collar and angular information ([2], Section 3)	Replaced by a one-line consequence of the area law (Lemma VIII.5).
Information reset by Gaussian filter ([2], Lemma 3.3)	Kept; one-dimensional locality (Section VIII B).
Adaptive patch constraints, encoded holes ([2], Sections 4, 6.1)	Unnecessary: no three owners meet on a line; logarithmic blocks are polynomial-dimensional registers.
Births, deaths, exchanges ([2], Lemmas 6.6–6.8)	Kept verbatim (Section IX).
Dyadic hierarchy with edge bands and point treatments ([2], Section 7)	Replaced by a linear sweep (Section X).
Density compression, column extraction ([2], Sections 5, 8)	Kept verbatim (Section XI).
Row–column routing with constant congestion	Replaced by placement on the line (Section XII).

TABLE I. What survives, what simplifies, what disappears.

Theorem I.1 (One-dimensional area law)**.** *There is* $C_{\mathrm{AL}} = C_{\mathrm{AL}}(q, R, J, \Delta)$*, with*

$$C_{\mathrm{AL}} \le C_0(q, R)\left(1 + \frac{J}{\Delta}\right),$$

such that for every finite $\Lambda \subset \mathbb{Z}$*, every admissible Hamiltonian satisfying* (2)*, and every* $A \subseteq \Lambda$*,*

$$S_\Omega(A) \le C_{\mathrm{AL}}\, |\partial_\Lambda A|.$$

Theorem I.2 (One-dimensional matrix product state approximation)**.** *Fix* q, R, J, Δ*. There are* $C, c > 0$ *depending only on these parameters such that for every* $L \ge 2$*, every admissible Hamiltonian on* $\Lambda = \{1, \dots, L\}$ *satisfying* (2)*, there is a nonzero matrix product state* Φ *on* Λ *with open boundary conditions and bond dimension at most* CL^c *such that*

$$\min_{\theta \in \mathbb{R}} \left\| \frac{\Phi}{\|\Phi\|} - \mathrm{e}^{i\theta}\Omega \right\| \le L^{-1}.$$

The constants never depend on Λ, on A, or on the particular Hamiltonian. No gap for a restricted Hamiltonian, no frustration freeness, no translation invariance is assumed.

a. Formalization. Theorems I.1 and I.2, with explicit constants, are formalized in Lean 4 in the TNLean modules `TNLean/MPS/AreaLaw/` (blueprint chapter "The one-dimensional area law and matrix product state approximation"). The formal constants differ from those printed here in the count of split interactions and in the shell width, as recorded in `docs/formalization/one-dimensional-chain.md`; Theorem I.2 is formalized by the two-line route of Section VII only, and the protocol route of Part II is not formalized.

Part I
The one-dimensional area law

II. SETTING, REDUCTIONS, AND COUNTING

In words. Three small reductions make the rest clean. If the domain Λ has several components, the Hamiltonian is a sum over components (every support has finite graph diameter, so it lies in one component), the unique ground vector is a product over components, and entropy and edge boundary both add over components. So we may take Λ to be an interval. A subset A of an interval is a union of maximal subintervals, each of which contributes at least one boundary edge, and subadditivity reduces the theorem to intervals. Finally, the only two counting facts we need are these: an interval is cut at two points at most, and an interaction of diameter at most R that crosses a cut lives in a window of $2R$ sites around that cut. So the number of interactions crossing the boundary of an interval, and the

logarithm of their joint dimension, are bounded by constants depending only on R and q. This is the entire "area" input of the one-dimensional proof.

Precisely. For an interval $X \subseteq \Lambda$ and a support Y with $\operatorname{diam}_{d_\Lambda} Y \le R$, say that Y is *split* by X if $Y \cap X \neq \varnothing$ and $Y \setminus X \neq \varnothing$. Since Y is contained in $R+1$ consecutive integers and contains a point on each side of one of the two cut edges of X, it lies in one of the two windows of $2R$ consecutive sites centered at a cut edge. Hence

$$\begin{aligned} &\#\{Y:\ Y \text{ split by } X\} \le 2 \cdot 2^{2R} =: N_R, \\ &\dim \mathcal{H}_Y \le q^{R+1}, \qquad \ell_R := \log(\mathrm{e} q^{R+1}), \end{aligned} \tag{3}$$

and a term h_Y split by X can be expanded by matrix units on $Y \cap X$ as

$$h_Y = \sum_{\alpha=1}^{N_Y} c_\alpha \otimes d_\alpha, \qquad N_Y \le q^{2(R+1)}, \qquad \|c_\alpha\|\,\|d_\alpha\| \le J, \tag{4}$$

with c_α supported in X and d_α in $\Lambda \setminus X$. (Each matrix unit E_{ab} on $\mathcal{H}_{Y\cap X}$ has norm one, and the coefficient operator $d_\alpha = \operatorname{Tr}_{Y\cap X}[(E_{ba}\otimes \mathrm{I}) h_Y]$ has norm at most $\|h_Y\|$.) Put

$$\mathcal{B}_R := 1 + N_R\, \ell_R^2. \tag{5}$$

Lemma II.1 (Reductions)**.** *It suffices to prove Theorem I.1 for Λ an interval and A an interval with $\partial_\Lambda A \neq \varnothing$, with a constant C_{int} bounding $S_\Omega(A)$; then Theorem I.1 holds with $C_{\mathrm{AL}} = C_{\mathrm{int}}$.*

Proof. Let $\Lambda_1, \dots, \Lambda_k$ be the components of Λ (maximal intervals). Every support of finite graph diameter lies in one component, so $H = \sum_i H_i$ with H_i acting on $\mathcal{H}_{\Lambda_i}$. The ground eigenspace of H is the tensor product of the ground eigenspaces of the H_i; if the former is one-dimensional so is each latter, hence $\Omega = \bigotimes_i \Omega_i$ and $S_\Omega(A) = \sum_i S_{\Omega_i}(A \cap \Lambda_i)$, while $\partial_\Lambda A = \bigsqcup_i \partial_{\Lambda_i}(A \cap \Lambda_i)$. Each H_i inherits (2) with the same Δ (restrict the inequality to vectors of the form $\xi \otimes \bigotimes_{j\neq i} \Omega_j$). So assume Λ is an interval. Write $A = \bigsqcup_{a=1}^m A_a$ as a union of maximal subintervals. If some $A_a = \Lambda$ then $A = \Lambda$ and $S(A) = 0$. Otherwise each A_a has at least one boundary edge, and distinct A_a have distinct boundary edges (two maximal subintervals are separated by at least one site, so their outer cut edges differ). Thus $m \le |\partial_\Lambda A|$ and subadditivity gives $S(A) \le \sum_a S(A_a) \le C_{\mathrm{int}} |\partial_\Lambda A|$. □

From now on $\Lambda = \{1, \dots, N\}$ and $X \subseteq \Lambda$ is an interval. For an integer $j \ge 0$ write $X^{+j} = \{x \in \Lambda : \operatorname{dist}(x, X) \le j\}$, again an interval.

III. THE MARGINAL TAIL BOUND

In words. Fix an interval X and write $\Omega = \sum_j \sqrt{p_j}\, u_j \otimes w_j$ for its Schmidt decomposition across X and the rest. The entanglement spectrum is the probability law of the *surprisal* $K = -\log p_j$ drawn with probability p_j; its mean is $S(X)$. The lemma says that, because of the global gap, K has exponential tails around its mean with a width that depends only on $q, R, J/\Delta$: the spectrum of an interval is *concentrated*, whatever its entropy is.

The mechanism is a tilt. Multiply Ω by $\rho_X^{-u/2}$ (acting on X) and renormalize: this reweights the Schmidt probabilities to $p_j^{1-u}/M(u)$, with $M(u) = \sum_j p_j^{1-u}$ the moment generating function of K. The tilted vector Θ is an exact eigenvector of the conjugated Hamiltonian FHF^{-1}. Interactions inside X or inside its complement do not see the conjugation in expectation (they commute with the marginal, or with F). Only the boundedly many interactions crossing a cut of X do, and for each of them the conjugated expectation $f(z) = \langle \Theta, \sigma^z h \sigma^{-z} \Theta\rangle$ is an entire function of the tilt parameter which is bounded on a strip and satisfies $f(-\bar z) = \overline{f(z)}$. That symmetry kills the linear term of the *real part*, so the excitation energy of Θ is quadratic in u. The gap converts a quadratic energy into a quadratic loss of overlap with Ω; that overlap is the ratio $M(u/2)^2/M(u)$. Iterating the resulting inequality $\log M(u) \le 2 \log M(u/2) + Cu^2$ down to $u \to 0$, where $\log M$ is linear with slope $S(X)$, gives $\log M(u) \le u S(X) + Cu^2$: a sub-Gaussian moment bound. Exponential Markov turns it into the tail bound.

In two dimensions the width of the tail grows like the square root of the boundary. In one dimension it is a constant. This constant is what makes both halves of this note work.

Precisely. Let $X \subseteq \Lambda$ be an interval, $\rho = \rho_X$ with positive eigenvalues p_j, let K be the random variable taking the value $-\log p_j$ with probability p_j, so that $\mathbb{E}K = S(X)$, and put

$$\vartheta = \frac{J}{\Delta}, \qquad u_0 = \frac{1}{32\sqrt{(1+\vartheta)\mathcal{B}_R}}. \tag{6}$$

Lemma III.1 (Marginal tail bound)**.** *For every interval $X \subseteq \Lambda$ and every real u with $|u| \le u_0$,*

$$\log \operatorname{Tr} \rho_X^{1-u} = \log \mathbb{E}\, \mathrm{e}^{uK} \le u\, S(X) + 512\, \mathrm{e}\, \vartheta\, \mathcal{B}_R\, u^2, \tag{7}$$

where ρ_X^{1-u} is zero on the kernel of ρ_X. Consequently, for every $w \ge 0$,

$$\Pr\{|K - S(X)| > w\} \le 2\mathrm{e}^{\mathrm{e}/2} \exp(-u_0 w). \tag{8}$$

Proof. Put $M(u) = \sum_j p_j^{1-u}$, $L(u) = \log M(u)$; note $L(0) = 0$ and $L'(0) = S(X)$. Let P be the support projection of ρ, let $F = \rho^{-u/2}$ on $P\mathcal{H}_X$ and $F = \mathrm{I}$ on the kernel, and define

$$\Theta = \frac{F\Omega}{\sqrt{M(u)}}, \qquad \sigma = \operatorname{Tr}_{\Lambda\setminus X} |\Theta\rangle\langle\Theta| = \frac{\rho^{1-u}}{M(u)}.$$

F is invertible on $\mathcal{H}_\Lambda$, so $FHF^{-1}\Theta = E_0\Theta$ and

$$\langle \Theta, (H - E_0)\Theta\rangle = \sum_Y \big(\langle \Theta, h_Y \Theta\rangle - \langle \Theta, F h_Y F^{-1} \Theta\rangle \big). \tag{9}$$

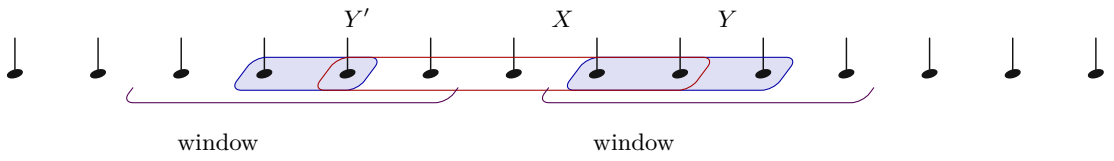


FIG. 1. An interval X of the chain (drawn for $R = 2$), the two windows of $2R$ sites centred on its cut edges, and two interaction supports Y' and Y split by X, one in each window. Every split support lies in a window, which is why their number is bounded by N_R.

A term supported in $\Lambda \setminus X$ commutes with F and contributes zero. A term supported in X contributes zero because $\operatorname{Tr}\sigma F h_Y F^{-1} = \operatorname{Tr}\sigma h_Y$ by $[F,\sigma]=0$ and cyclicity. So only terms split by X remain, and there are at most N_R of them by (3).

Fix such a term $h = h_Y$. For complex z let $\sigma^z = P\exp(z\log(\sigma|_{P\mathcal{H}_X}))P$ and $f(z) = \langle\Theta, \sigma^z h \sigma^{-z}\Theta\rangle$, an entire function with $f(0) = \langle\Theta, h\Theta\rangle$ (because Θ lies in the support of σ). At $z_u = -u/(2(1-u))$ one has, on the support, $\sigma^{z_u} = M(u)^{-z_u}F$ and $\sigma^{-z_u} = M(u)^{z_u}F^{-1}$, so $f(z_u) = \langle\Theta, FhF^{-1}\Theta\rangle$. On the imaginary axis the powers are unitary on the support and $|f(iy)| \le J$. For $|\Re z| \le \frac14$ expand h as in (4) and let $\Theta = \sum_j \sqrt{s_j}\, e_j \otimes f_j$ be its Schmidt decomposition across X. Then

$$f(z) = \sum_\alpha \sum_{j,k} s_j^{1/2+z} s_k^{1/2-z}\, \langle e_j, c_\alpha e_k\rangle\langle f_j, d_\alpha f_k\rangle.$$

For $|\Re z| \le \frac14$, $|s_j^{1/2+z}s_k^{1/2-z}| \le s_j + s_k$ by the weighted arithmetic–geometric mean inequality, and for any operators c, d the row and column Cauchy–Schwarz inequalities give

$$\sum_{j,k} (s_j + s_k)\,|c_{jk}|\,|d_{jk}| \le 2\|c\|\,\|d\|, \tag{10}$$

because $\sum_k |c_{jk}||d_{jk}| \le \|c\|\|d\|$ for each row j and likewise for each column. Hence $|f(z)| \le 2Jq^{2(R+1)}$ on $|\Re z| \le \frac14$. Hadamard's three-lines theorem on each of the strips $0 \le \pm\Re z \le \frac14$ gives $\log|f(x+iy)| \le \log J + 4|x|\log(2q^{2(R+1)}) \le \log J + 8|x|\,\ell_R$, so that

$$|f(z)| \le \mathrm{e}J \qquad \text{for } |\Re z| \le \delta_R := \frac{1}{8\ell_R}. \tag{11}$$

Since $h = h^*$ and $(\sigma^z)^* = \sigma^{\bar z}$, one has $f(-\bar z) = \overline{f(z)}$. Writing $f(z) = \sum_n c_n z^n$, this says c_n is real for even n and purely imaginary for odd n; so for real x, $\Re(f(x) - f(0)) = \sum_{n\ge2 \text{ even}} c_n x^n$, and Cauchy's estimate $|c_n| \le \mathrm{e}J\delta_R^{-n}$ from (11) gives, for $|x| \le \delta_R/2$,

$$|\Re(f(x) - f(0))| \le 2\mathrm{e}J\, x^2\delta_R^{-2} = 128\,\mathrm{e}J\,\ell_R^2\,x^2. \tag{12}$$

For $|u| \le u_0$ we have $|u| \le \frac{1}{32}$, $|z_u| \le |u|$ and $\ell_R \le \sqrt{\mathcal{B}_R}$, so $|z_u| \le \delta_R/2$. Since (9) is real, applying (12) to each of the at most N_R split terms yields

$$0 \le \langle\Theta, (H - E_0)\Theta\rangle \le 128\,\mathrm{e}J\,N_R\ell_R^2\,u^2 \le 128\,\mathrm{e}J\,\mathcal{B}_R\,u^2. \tag{13}$$

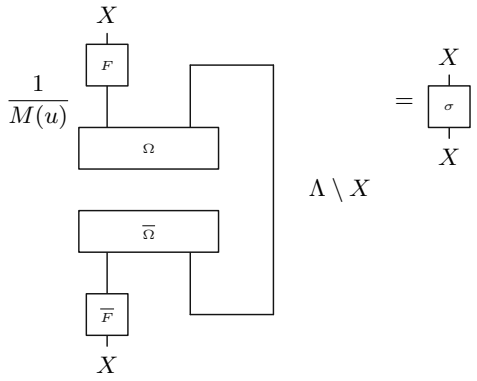


FIG. 2. The tilt as a tensor diagram. Multiplying Ω on the interval X by $F = \rho_X^{-u/2}$ and tracing out the rest of the chain gives $\sigma = \rho_X^{1-u}/M(u)$. Only interactions crossing the cuts of X fail to commute with F, so the tilted state has excitation energy of order u^2.

The gap (2) now gives $1 - |\langle\Omega,\Theta\rangle|^2 \le 128\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2 \le \frac12$, and $\langle\Omega,\Theta\rangle = M(u/2)/\sqrt{M(u)}$, so

$$\frac{M(u/2)^2}{M(u)} \ge 1 - 128\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2, \qquad \text{i.e.}$$
$$L(u) \le 2L(u/2) + 256\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2,$$

using $-\log(1-v) \le 2v$ for $0 \le v \le \frac12$. Every halving $u/2^j$ stays in the admissible interval, so by induction

$$L(u) \le 2^j L(u/2^j) + 256\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2 \sum_{i<j} 2^{-i}$$
$$\xrightarrow[j\to\infty]{} uS(X) + 512\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2,$$

which is (7). For (8), apply Markov's inequality to $\mathrm{e}^{\pm u_0(K-S)}$: the quadratic term at $u = \pm u_0$ equals $512\mathrm{e}\vartheta\mathcal{B}_R/(1024(1+\vartheta)\mathcal{B}_R) \le \mathrm{e}/2$, so each one-sided tail is at most $\mathrm{e}^{\mathrm{e}/2}\mathrm{e}^{-u_0 w}$. □

Remark III.2 (Rényi entropies near one)**.** For $0 < u \le u_0$ and $\alpha = 1-u$, (7) reads $S_\alpha(X) = \frac{1}{u}\log\operatorname{Tr}\rho_X^\alpha \le S(X) + 512\mathrm{e}\vartheta\mathcal{B}_R(1-\alpha)$. Since $S_\alpha \ge S$ for $\alpha < 1$, all Rényi entropies of order $\alpha \in [1-u_0, 1]$ of an interval lie within $16\mathrm{e}\sqrt{(1+\vartheta)\mathcal{B}_R}$ of its von Neumann entropy. This is used in Section VII.

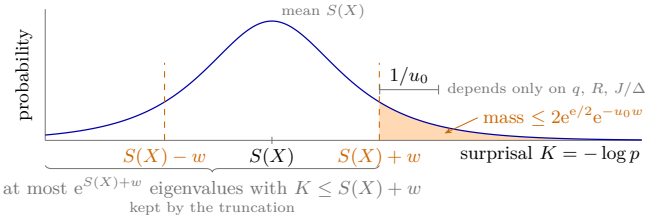


FIG. 3. Schematic law of the surprisal. The tails decay exponentially at rate u_0; the mass beyond $S(X)+w$ is at most $2e^{e/2}e^{-u_0 w}$, and at most $e^{S(X)+w}$ eigenvalues lie to the left of that line. Both facts are reused for the sequential truncation in Part II.

IV. THE BUFFERED DISCOUNT WITH A CONSTANT-WIDTH BUFFER

In words. Here is the one lemma that does the work. Take an interval X and surround it by m nested shells: $X_j = X^{+jD}$ for $j = 1, \dots, m$, each one step of width $D = R$ thicker than the previous one, and let $T_j = X_j \setminus X$ be the two flanks. The number m will be a large constant (of order $1 + J/\Delta$), so the total buffer $T = T_m$ has *constant* width mD on each side of X. The claim is

$$S(X\,|\,T) \;\le\; \tfrac{1}{2}\,S(X) + C.$$

Conditioning X on a constant number of neighboring sites halves its entropy, up to a constant.

Why should that be? Apply to Ω a product $K = L_m \cdots L_1$ of positive operators, L_j acting on X_j, each normalized by $\mathrm{Tr}\, L_j^{2/a_j} = 1$ with $a_j = 2/m$ (so the weights sum to 2), and maximize the norm $N = \|K\Omega\|$. Two estimates bound the optimum from opposite sides.

The optimum is small. At the optimum, each filter commutes with the marginal of the filtered vector on its own region (a stationarity computation), which lets one strip the outer filters from any expectation of an operator living inside them. Then the energy of the normalized filtered vector is at most a constant times $\sum_j a_j^2 \times$ (number of interactions split by X_j): an interaction is split by at most one shell because consecutive shells are $D \ge R$ apart, and the conjugation of a split term by one filter changes its expectation only quadratically in a_j, by the same kind of row–column Cauchy–Schwarz estimate as in the tail bound. In one dimension the number of split terms is bounded, so the energy is at most C/m. The gap then says the filtered vector overlaps Ω to $1 - C/(m\Delta)$. Using that overlap, one peels the filters off one at a time, and each peeled filter L_j can shrink the norm by at most $\|L_j\Omega\| \le \exp(-a_j S(X_j)/2 + C a_j^2)$ by Hölder and the tail bound. So $-2\log N \ge \sum_j a_j S(X_j) - C$.

The optimum is large. Pin a Schmidt vector u_i of X, that is, use the trial filters $L_j = |u_i\rangle\langle u_i| \otimes M_j$ with M_j a filter on the flank T_j only. The output is $\sqrt{p_i}$ times a filtered version of the partner vector w_i, and an interpolation argument shows the flank filters can be chosen so that the norm is at least $\exp(-\sum_j a_j S_{w_i}(T_j)/2)$. Averaging over i and using concavity of entropy, $-2\log N \le S(X) + \sum_j a_j S(T_j)$.

Subtracting, $\sum_j a_j\,[S(X_j) - S(T_j)] \le S(X) + C$, and $S(X_j) - S(T_j)$ is the conditional entropy $S(X|T_j)$, which decreases as the flank grows. With $\sum_j a_j = 2$ this is the discount. Notice what did not happen: no property of the state on the flanks was used beyond their being a constant number of sites, and no gap of any restricted Hamiltonian was invoked.

Precisely. Fix $D = R$ (recall $R \ge 1$) and an integer $m \ge 4$. For an interval $X \subseteq \Lambda$ put

$$X_0 = X, \qquad X_j = X^{+jD}\ (1 \le j \le m),$$
$$T_j = X_j \setminus X, \qquad a_j = \frac{2}{m}.$$

Then $T_1 \subseteq \cdots \subseteq T_m$, each T_j consists of at most two intervals of length at most jD, and $\sum_j a_j = 2$. Define the constants

$$C_E := 8\,N_R\,J\,q^{2(R+1)},$$
$$m := \Big\lceil \max\Big\{4,\ \frac{4C_E}{\Delta},\ 128\sqrt{(1+\vartheta)\mathcal{B}_R},\ 4096\,\mathrm{e}\,(1+\vartheta)\,\mathcal{B}_R\Big\}\Big\rceil. \tag{14}$$

Every entry is at most a constant depending on q, R times $(1 + J/\Delta)$, so

$$m \le C_1(q,R)\,(1+\vartheta). \tag{15}$$

Lemma IV.1 (Single splitting)**.** *A support Y with $\mathrm{diam}_{d_\Lambda} Y \le R$ is split by at most one of the intervals $X_1, \dots, X_m$. If it is split by X_j, it is contained in X_i for every $i > j$ and disjoint from X_i for every $i < j$. The same holds verbatim for the family $X_0, \dots, X_m$.*

Proof. Suppose Y is split by X_j. It contains a point $y \notin X_j$ and a point $x \in X_j$, and all its points are within graph distance $R \le D$ of x and of y. Points within distance D of X_j lie in X_{j+1}, so $Y \subseteq X_{j+1} \subseteq X_i$ for $i > j$. Points of X_{j-1} are at distance more than D from every point outside X_j (because $X_j = X_{j-1}^{+D}$), so $Y \cap X_{j-1} = \varnothing$, and likewise for $i < j$. A set contained in X_i or disjoint from X_i is not split by X_i. □

Lemma IV.2 (Buffered discount)**.** *With m as in (14), every interval $X \subseteq \Lambda$ satisfies*

$$S(X\,|\,T_m) \;\le\; \frac{1}{2}\,S(X) + \frac{1}{2} + \frac{C_E}{\Delta}. \tag{16}$$

Proof. If $X = \varnothing$ both sides vanish. Assume $X \ne \varnothing$.

a. The optimization. Fix $\epsilon > 0$ so small that $(\dim \mathcal{H}_{X_j})\epsilon^{2/a_j} < 1$ for all j, and maximize

$$N = \|L_m \cdots L_1 \Omega\|$$

over positive operators L_j on $\mathcal{H}_{X_j}$ (extended by the identity) with $L_j \ge \epsilon \mathrm{I}$ and $\mathrm{Tr}_{\mathcal{H}_{X_j}} L_j^{2/a_j} = 1$. The feasible sets are nonempty and compact, every feasible product is invertible, so a maximum $N > 0$ exists. At an optimum put $K = L_m \cdots L_1$, $\phi = K\Omega/N$, and $\rho_j = \mathrm{Tr}_{\Lambda \setminus X_j} |\phi\rangle\langle\phi|$.

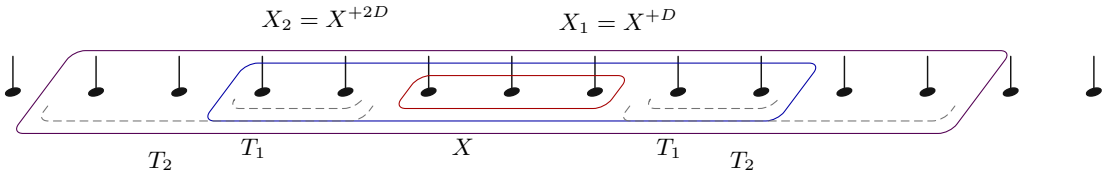


FIG. 4. The interval X and the nested shells $X_1 = X^{+D} \subset X_2 = X^{+2D}$, drawn with $D = R = 2$; the braces mark the flanks $T_j = X_j \setminus X$, each a union of at most two intervals of length at most jD, which carry the buffer.

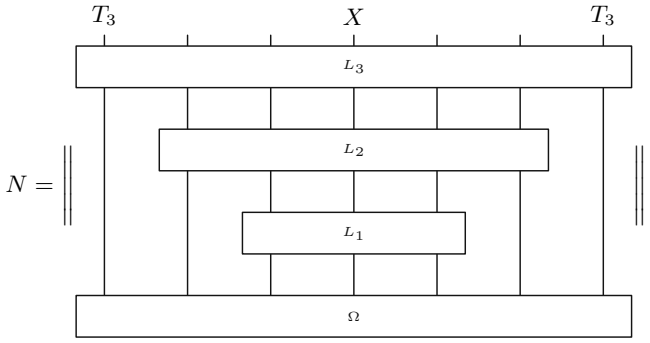


FIG. 5. The optimized norm. Three positive filters L_1, L_2, L_3 act on the nested intervals $X_1 \subset X_2 \subset X_3$ of the ground vector; each is normalized by $\operatorname{Tr} L_j^{2/a_j} = 1$.

b. Stationarity. We claim $[L_j, \rho_j] = 0$ for all j, by descending induction on j. The tool is the stripping identity: if $[L_k, \rho_k] = 0$ and T is supported in X_k, then

$$\langle \phi, L_k T L_k^{-1} \phi \rangle = \operatorname{Tr}(L_k^{-1} \rho_k L_k T) = \operatorname{Tr}(\rho_k T) = \langle \phi, T\phi \rangle. \tag{17}$$

Assume the claim for $k > j$ and let B be anti-Hermitian on $\mathcal{H}_{X_j}$. Replacing L_j by $\mathrm{e}^{tB} L_j \mathrm{e}^{-tB}$ preserves both constraints, so the derivative of $\log\|K\Omega\|$ at $t = 0$ vanishes. That derivative equals $\Re\langle \phi, K_{>j}(B - L_j B L_j^{-1}) K_{>j}^{-1} \phi \rangle$ with $K_{>j} = L_m \cdots L_{j+1}$. The operator inside is supported in X_j, hence in X_m; strip L_m by (17), then L_{m-1}, and so on down to L_{j+1}. Since $\operatorname{Tr}(\rho_j B)$ is purely imaginary, stationarity gives $\Re \operatorname{Tr}(L_j^{-1} \rho_j L_j B) = 0$ for all anti-Hermitian B, so $L_j^{-1} \rho_j L_j$ is Hermitian, i.e. equals $L_j \rho_j L_j^{-1}$, so ρ_j commutes with L_j^2 and hence with $L_j \geq 0$. No commutation between distinct filters is claimed or used.

In a common eigenbasis let l_i, p_i be the eigenvalues of L_j, ρ_j, and $x_i = l_i^{2/a_j}$, so $\sum_i x_i = 1$, $x_i \geq \epsilon^{2/a_j}$. A diagonal variation and the same stripping give $d \log N = \frac{a_j}{2} \sum_i p_i \, dx_i / x_i$. The Kuhn–Tucker conditions on the simplex with the floor say that p_i/x_i is a common value $\lambda > 0$ on coordinates above the floor and at most λ on floored coordinates (there is a free coordinate by the dimensional slack, and $\lambda > 0$ because moving mass from a zero-probability free coordinate to a positive-probability floored one would increase N). Hence $x_i = \max\{\epsilon^{2/a_j}, p_i/\lambda\}$ and, since taking a maximum with a common floor contracts differences of logarithms,

$$\left| \log \frac{l_i}{l_k} \right| \leq \frac{a_j}{2} \left| \log \frac{p_i}{p_k} \right| \qquad (p_i, p_k > 0). \tag{18}$$

c. Energy of the optimized vector. Since $KHK^{-1}\phi = E_0 \phi$,

$$\langle \phi, (H - E_0)\phi \rangle = \sum_Y \Re\big(\langle \phi, h_Y \phi \rangle - \langle \phi, K h_Y K^{-1} \phi \rangle \big).$$

If Y is split by no X_j ($1 \leq j \leq m$), then by Lemma IV.1 each X_j either contains Y or is disjoint from it; strip the containing filters by (17) from the outside in, and commute the disjoint ones through h_Y: the term vanishes. If Y is split by X_j, the same procedure leaves exactly $\langle \phi, h_Y \phi \rangle - \langle \phi, L_j h_Y L_j^{-1} \phi \rangle$. Let $\phi = \sum_i \sqrt{p_i}\, e_i \otimes f_i$ be the Schmidt decomposition across X_j, with the e_i chosen in the common eigenbasis of ρ_j and L_j (within each eigenspace of ρ_j one may diagonalize L_j, since they commute), and $h_{ik} = \langle e_i \otimes f_i, h_Y (e_k \otimes f_k) \rangle$. Pairing (i,k) with (k,i) and using $h_{ki} = \overline{h_{ik}}$,

$$\begin{aligned} &\Re\langle \phi, L_j h_Y L_j^{-1} \phi \rangle - \langle \phi, h_Y \phi \rangle \\ &\qquad = \sum_{i,k:\, p_i p_k > 0} \sqrt{p_i p_k} \, \big(\cosh(\log(l_i/l_k)) - 1 \big) \Re h_{ik}. \end{aligned}$$

Put $z = \frac{1}{2} \log(p_i/p_k)$. By (18) and $a_j \leq \frac{1}{2}$, $\cosh(\log(l_i/l_k)) - 1 \leq \cosh(a_j z) - 1 \leq \frac{1}{2} a_j^2 z^2 \cosh(a_j z) \leq \frac{1}{2} a_j^2 z^2 \cosh(z/2)$, while $\sqrt{p_i p_k} = (p_i + p_k)/(2\cosh z)$. Since $\sup_z z^2 \cosh(z/2)/(4 \cosh z) \leq 1$, the coefficient is at most $a_j^2 (p_i + p_k)$. Expanding h_Y by (4) and using (10)

for each product term gives

$$\left|\Re\langle\phi, L_j h_Y L_j^{-1}\phi\rangle - \langle\phi, h_Y\phi\rangle\right| \le 2a_j^2\, q^{2(R+1)} J.$$

Summing over the at most N_R terms split by each X_j,

$$\begin{aligned} 0 \le \langle\phi, (H-E_0)\phi\rangle &\le \sum_{j=1}^{m} N_R \cdot 2a_j^2 q^{2(R+1)} J \\ &= \frac{8N_R J q^{2(R+1)}}{m} = \frac{C_E}{m}. \end{aligned} \tag{19}$$

By (2) and (14),

$$1 - |\langle\Omega, \phi\rangle|^2 \le \frac{C_E}{m\Delta} \le \frac{1}{4}. \tag{20}$$

The same bound holds for the optimizer of every prefix $L_j \cdots L_1$, $1 \le j \le m$: the stationarity and energy arguments apply verbatim to the prefix problem with the same weights and floor, and a term split by some X_k with $k > j$ is disjoint from all X_i, $i \le j$, so it contributes nothing; the prefix energy is bounded by the sum in (19) restricted to $i \le j$.

d. Upper bound on the optimum. Let N_j be the optimum for the prefix $1, \dots, j$ at the same floor, $N_0 = 1$, and let $L_j \cdots L_1$ be an optimizer for prefix j with $K_{<j} = L_{j-1} \cdots L_1$. By (20) for that prefix and Cauchy–Schwarz,

$$\begin{aligned} \sqrt{1 - C_E/(m\Delta)}\; N_j &\le |\langle\Omega, L_j K_{<j}\Omega\rangle| \\ &= |\langle L_j\Omega, K_{<j}\Omega\rangle| \le \|L_j\Omega\|\, N_{j-1}. \end{aligned}$$

For any feasible L_j, Schatten–Hölder with exponents $\frac{1}{1-a_j}$ and $\frac{1}{a_j}$ and the constraint $\operatorname{Tr} L_j^{2/a_j} = 1$ give

$$\|L_j\Omega\|^2 = \operatorname{Tr}\rho_{X_j} L_j^2 \le \left(\operatorname{Tr}\rho_{X_j}^{1/(1-a_j)}\right)^{1-a_j}.$$

Apply Lemma III.1 to the interval X_j with $u = -a_j/(1-a_j)$, which is admissible because $|u| \le 2a_j = 4/m \le u_0$ by (14). Using $(1-a_j)u^2 \le 2a_j^2$,

$$\log\|L_j\Omega\|^2 \le -a_j S(X_j) + 1024\,\mathrm{e}\,\vartheta\,\mathcal{B}_R\, a_j^2.$$

Multiplying the recursion over j and using $-\frac{1}{2}\log(1-v) \le v$ for $v \le \frac{1}{2}$,

$$\begin{aligned} \log N &\le -\frac{1}{2}\sum_j a_j S(X_j) + 512\,\mathrm{e}\,\vartheta\,\mathcal{B}_R \sum_j a_j^2 + \frac{mC_E}{m\Delta} \\ &\le -\frac{1}{2}\sum_j a_j S(X_j) + \frac{1}{2} + \frac{C_E}{\Delta}, \end{aligned}$$

because $\sum_j a_j^2 = 4/m \le 1/(1024\mathrm{e}\vartheta\mathcal{B}_R)$ by (14). That is,

$$-2\log N \;\ge\; \sum_{j=1}^{m} a_j S(X_j) - 1 - \frac{2C_E}{\Delta}. \tag{21}$$

The constants in (21) do not depend on the floor ϵ.

e. Lower bound on the optimum by Schmidt pinning. Let $\epsilon \downarrow 0$. The optima converge to the optimum N over nonnegative filters with the same trace constraints (the feasible sets are compact and every nonnegative feasible operator is a limit of strictly positive feasible ones), and (21) persists. Write $\Omega = \sum_i \sqrt{p_i}\, u_i \otimes w_i$ with $p_i > 0$ for the Schmidt decomposition across X. For fixed i use the trial filters

$$\begin{aligned} &L_j = |u_i\rangle\langle u_i| \otimes M_j, \qquad M_j \ge 0 \text{ on } \mathcal{H}_{T_j}, \\ &\operatorname{Tr} M_j^{2/a_j} = 1, \end{aligned}$$

which are feasible (as $\operatorname{Tr}_{X_j} L_j^{2/a_j} = \operatorname{Tr} M_j^{2/a_j} = 1$) and give $N \ge \sqrt{p_i}\, N_i$, where $N_i = \sup\|M_m \cdots M_1 w_i\|$ over such M_j. We claim

$$\log N_i \ge -\frac{1}{2}\sum_j a_j S_{w_i}(T_j). \tag{22}$$

Fix invertible feasible M_j and consider $F_i(z) = M_m^z \cdots M_1^z w_i$ on $0 \le \Re z \le 1$, bounded and analytic. On $\Re z = 0$ the factors are unitary and $\|F_i(iy)\| = 1$. On $\Re z = 1$ put $U_j = M_j^{iy}$, $V_j = U_j \cdots U_1$, $V_0 = \mathrm{I}$; moving the unitaries left,

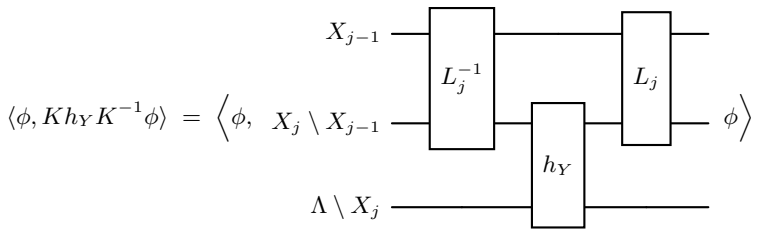
$$\begin{aligned} &M_m^{1+iy} \cdots M_1^{1+iy} \\ &\quad = (U_m \cdots U_1)\,(V_{m-1}^* M_m V_{m-1}) \cdots (V_0^* M_1 V_0). \end{aligned}$$

Each conjugated factor is positive, supported in T_j (the conjugating unitaries live on the smaller nested flanks), and has the same trace-power normalization, so $\|F_i(1+iy)\| \le N_i$. Testing against unit vectors and applying the scalar three-lines theorem gives $\log\|F_i(x)\| \le x\log N_i$ for $0 \le x \le 1$; differentiating at $x = 0^+$ yields $\sum_j \langle w_i, \log M_j\, w_i\rangle \le \log N_i$. Taking $M_j = \big((1-\delta)\rho_{w_i,T_j} + \delta \mathrm{I}/\dim\mathcal{H}_{T_j}\big)^{a_j/2}$ and letting $\delta \downarrow 0$, the left side tends to $-\frac{1}{2}\sum_j a_j S_{w_i}(T_j)$, proving (22). Hence $-2\log N \le -\log p_i + \sum_j a_j S_{w_i}(T_j)$ for every i; average with weights p_i and use concavity of the entropy (the p_i-average of ρ_{w_i,T_j} is ρ_{T_j}):

$$-2\log N \le S(X) + \sum_{j=1}^{m} a_j S(T_j). \tag{23}$$

f. Conclusion. Subtract (23) from (21): $\sum_j a_j\big(S(X_j) - S(T_j)\big) \le S(X) + 1 + 2C_E/\Delta$. Here $S(X_j) - S(T_j) = S(XT_j) - S(T_j) = S(X|T_j)$, and $S(X|T_j) \ge S(X|T_m)$ for $j \le m$ by strong subadditivity (conditioning on the larger flank $T_m \supseteq T_j$ decreases conditional entropy). With $\sum_j a_j = 2$ this gives $2S(X|T_m) \le S(X) + 1 + 2C_E/\Delta$. □

V. CLOSING THE ARGUMENT



outer filters $L_{j+1}, \dots, L_m$: stripped by the stripping identity, $[L_k, \rho_k] = 0$
inner filters $L_1, \dots, L_{j-1}$: disjoint from Y, commute with h_Y

FIG. 6. Why the energy is small. An interaction h_Y split by X_j is conjugated by exactly one filter: the outer filters, whose regions contain Y, are stripped by the stripping identity at the optimum, and the inner ones commute with h_Y because they live on disjoint sites.

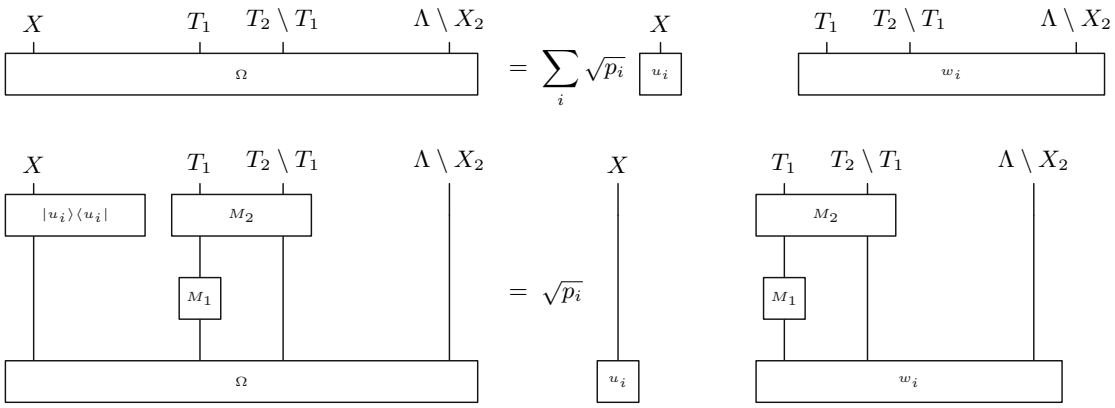


FIG. 7. Schmidt pinning. With the trial filters $|u_i\rangle\langle u_i| \otimes M_j$ (the projector is drawn once, by idempotence; the flanks are grouped into one leg each), the filtered vector becomes $\sqrt{p_i}\, u_i$ tensored with the flank filters applied to the partner w_i, which lower-bounds the optimum by $-S(X) - \sum_j a_j S(T_j)$ in the exponent after averaging over i.

In words. The discount says that $S(X|T) \le \frac{1}{2}S(X) + C$ where T is a buffer of a constant number of sites. But a conditional entropy cannot be very negative: by the Araki–Lieb triangle inequality $S(XT) \ge S(X) - S(T)$, so $S(X|T) \ge S(X) - 2S(T)$. Putting the two together, $S(X) - 2S(T) \le \frac{1}{2}S(X) + C$, hence $S(X) \le 4S(T) + 2C$, and $S(T)$ is at most the number of sites of T times $\log q$. That is the area law.

Nothing else is needed. In two dimensions the same two inequalities give $S(X) \le 4S(T) + Cr$ for a box X of side r with a buffer T of width proportional to r; the buffer is an annulus of area $\sim r^2$ whose entropy is as unknown as that of X, and the inequality is useless. That is the precise point at which the two-dimensional proof must bring in everything else.

Precisely.

Proof of Theorem I.1. By Lemma II.1 it suffices to bound $S(X)$ for an interval X in an interval domain. Let $T = T_m$ be the buffer of Lemma IV.2; it has at most $2mD$ sites. The Araki–Lieb inequality $S(XT) \ge S(X) - S(T)$ gives $S(X|T) \ge S(X) - 2S(T)$, and (16) gives $S(X|T) \le \frac{1}{2}S(X) + \frac{1}{2} + C_E/\Delta$. Hence

$$S(X) \le 4S(T) + 1 + \frac{2C_E}{\Delta} \le 8mD \log q + 1 + \frac{2C_E}{\Delta} =: C_{\text{int}}. \tag{24}$$

By (15) and (14), $C_{\text{int}} \le C_0(q,R)(1 + J/\Delta)$ with, for instance, $C_0(q,R) = 8C_1(q,R)\, R \log q + 1 + 16 N_R q^{2(R+1)}$. □

Remark V.1 (What the constant depends on)**.** The gap enters only through $\vartheta = J/\Delta$, linearly. The dependence on the local dimension and range is through $N_R\, q^{2(R+1)}$, which for nearest-neighbor interactions ($R = 1$) is of order q^4. The constants were carried over from the two-dimensional source without optimization; Remark V.2 shows that the polynomial dependence on q is an artifact of one crude estimate. By comparison, the bound of Arad–Kitaev–Landau–Vazirani [4] is $\tilde{O}(\log^3(q)/\Delta)$ for $R = 1$.

Remark V.2 (Sharper constants, and what is intrinsic)**.** Three of the constants are not intrinsic to the argument.

The count N_R. The bound $2 \cdot 2^{2R}$ allows one interaction for every subset of a window. When the Hamiltonian has one term for each window of $R+1$ consecutive sites, a cut

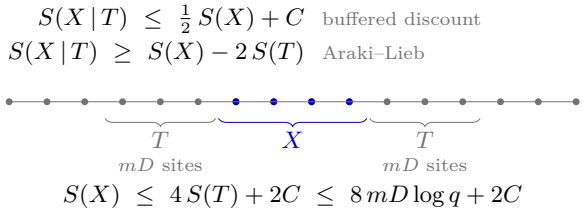


FIG. 8. The closing step. The interval X with its constant-width buffers T; the discount bounds $S(X|T)$ from above, ArakiLieb from below, and T has at most $2mD$ sites.

edge splits at most R of them, and N_R may be replaced by $2R$.

The factor $q^{2(R+1)}$ in C_E. It enters in (19) through the matrix-unit expansion (4) bounded term by term. The proof of Lemma III.1 met the same quantity and turned the dimension into a logarithm with the three-lines theorem; the same device works here. For a split term $h = h_Y$ and the optimal filter $L = L_j$ put $g(z) = \langle\phi, L^z h L^{-z}\phi\rangle$. It is entire, $g(0) = \langle\phi, h\phi\rangle$, $g(1) = \langle\phi, LhL^{-1}\phi\rangle$, $|g(iy)| \le J$ because L^{iy} is unitary, and $g(-\bar z) = \overline{g(z)}$. In the common eigenbasis, $g(z) = \sum_{i,k}\sqrt{p_i p_k}\,(l_i/l_k)^z h_{ik}$, and (18) with the weighted arithmetic–geometric mean inequality gives $\sqrt{p_i p_k}\,|(l_i/l_k)^z| \le p_i + p_k$ for $|\Re z| \le 1/a_j$; hence $|g(z)| \le 2Jq^{2(R+1)}$ there by (10). Since $\log(2q^{2(R+1)}) \le 2\ell_R$, the three-lines theorem gives $|g(z)| \le \mathrm{e}J$ for $|\Re z| \le 1/(2a_j\ell_R)$, and Cauchy's estimate as in (12) gives, because $a_j \le 1/(4\ell_R)$,

$$|\Re\big(g(1) - g(0)\big)| \le 8\,\mathrm{e}\,J\,\ell_R^2\,a_j^2.$$

So (19) holds with $2a_j^2 q^{2(R+1)}J$ replaced by $8\mathrm{e}J\ell_R^2 a_j^2$, that is, with

$$C_E' = 32\,\mathrm{e}\,N_R\,J\,\ell_R^2 \le 32\,\mathrm{e}\,J\,\mathcal{B}_R$$

in place of C_E. The entry $4C_E'/\Delta$ of (14) is then dominated by $4096\mathrm{e}(1+\vartheta)\mathcal{B}_R$, and (24) becomes

$$S(X) \;\le\; C\,(1 + J/\Delta)\,\mathcal{B}_R\,R\log q,$$

where now $\mathcal{B}_R \le 1 + 2R\big(1 + (R+1)\log q\big)^2$ and C is an absolute constant. For nearest-neighbor chains this is $O\big((1 + J/\Delta)\log^3 q\big)$, the order of the Arad–Kitaev–Landau–Vazirani bound [4], obtained without a Lieb–Robinson bound or an approximate ground-state projector. For range R it is $O\big((1+J/\Delta)R\log^3(q^R)\big)$, one factor R above the blocked nearest-neighbor bound.

What is intrinsic. Two features of the constants cannot be removed by sharper estimates within this route. First, the linear dependence on J/Δ: the Schmidt-pinning bound (23) carries $S(X)$ with coefficient one, so the closing step needs total weight $\sum_j a_j > 1$, and a total weight of order one spread over m shells costs the filtered vector an energy of order $J\mathcal{B}_R/m$ by (19); the overlap step (20) requires that energy to lie below the gap, hence $m \gtrsim J\mathcal{B}_R/\Delta$, and the buffer entropy $S(T) \le 2mR\log q$ is linear in J/Δ. Second, the width $u_0 \sim (\vartheta\mathcal{B}_R)^{-1/2}$ of the tail window in Lemma III.1: the tilted vector of (13) has energy $O(J\mathcal{B}_R u^2)$ for all $u \le 1/(16\ell_R)$, but the gap is used only through the variational inequality (2), which is vacuous once that energy exceeds Δ. The window width sets the rate of the Schmidt tail (8) and therefore the exponent of L in the bond dimension of Theorem I.2 (Section VII), which is of order $\sqrt{(1 + J/\Delta)\mathcal{B}_R}$. In the approximate-ground-state-projector route of [4] the gap-dependent quantities sit in a prefactor and the exponent of the chain length is universal. Moving $\sqrt{J/\Delta}$ out of that exponent within the present route would require using the gap in an amplified form, which is exactly what an approximate ground-state projector supplies.

Remark V.3 (What was used)**.** The proof used the global gap in exactly two places: in Lemma III.1 applied at the cuts of the shells X_j, and to convert the energy bound (19) into the overlap bound (20). No Lieb–Robinson bound, no spectral filter, no approximate ground-state projector, no Chebyshev polynomial, no bootstrapping of a Schmidt-rank bound appears. The entropy itself enters only through operator norms of filtered vectors.

VI. THE LONG ROUTE, FOR COMPARISON

In words. It is instructive to see what the two-dimensional proof would do in one dimension if one refused to use the Araki–Lieb shortcut, because this is where each piece of the big machine would be used, and because the amplification and two-family steps reappear, in spirit, in Part II.

Sub-volume exponent. Split an interval of length Mr into M children of length r and add them up in a random order. A child whose constant-width buffer has already been added gets the discount, which happens with a fixed probability p_0. This gives $F(Mr) \le (1 - p_0/4)M\,F(r) + CM$ for the maximal interval entropy F, hence $F(r) \le Cr^{e_0}$ with $e_0 < 1$. In two dimensions the analogous recursion gives r^{1+e_0}, which is *above* the boundary r; in one dimension r^{e_0} is already below the diameter r, which is all the next step needs.

Amplification. The amplification proposition of the source converts an information bound $I(X : \text{far}) \le n^{1-\varepsilon}$ into $I(X : J) \le n^{-M}$ for all J beyond a graph distance $2r$ with $r = O((n^{1-\varepsilon} + \log n)^{1/\alpha})$, $\alpha < 1$ as close to one as desired, using positive quasi-local constraints built from Kitaev's spectral

filter, a random product of the corresponding contractions, and a polar decomposition. In one dimension the trivial bound $I(X : \text{anything}) \le 2S(X) \le Cn^{e_0}$ for an interval X of length n already is such an input, with no collar at all; one gets $I(X : J) \le n^{-M}$ beyond distance $Cn^{e_0/\alpha} = o(n)$ once $\alpha > e_0$.

Two families. Cut the half-line to the right of the cut into dyadic intervals $X_i = [2^i, 2^{i+1})$ and alternate their family labels. Each X_i is separated from all earlier same-family intervals and from the other side of the cut by the gap X_{i-1}, of length $2^{i-1} \gg 2^{ie_0/\alpha}$. The two-family cancellation lemma then gives $S(A) \le |D| \log q + \frac{1}{2}\sum_i 2^{-iM}$ with $|D| = O(1)$.

So the long route also works, with the entire replica scanner replaced by the trivial inequality $I \le 2S$. We do not pursue it; the direct proof is shorter and gives better constants.

Precisely. We only record the recursion, since the other two steps are literally the source propositions with the stated inputs. Let $F(r)$ be the supremum of $S(X)$ over intervals $X \subseteq \Lambda$ of length at most r (over all admissible instances), and let $w_0 = mD$ be the buffer width of Lemma IV.2 with discount constant $C_{\text{buf}} = \frac{1}{2} + C_E/\Delta$. Let $M \ge 8(2w_0/r + 2)$ and let X be an interval of length at most Mr. Cut X into at most M consecutive children of length at most r and order them by a uniformly random permutation. Call a child interior if its buffer of width w_0 on each side lies in X; at most $B := 2\lceil w_0/r \rceil + 2$ children are not interior. For an interior child, the event that all children meeting its buffer precede it has probability at least $p_0 = (2\lceil w_0/r \rceil + 3)^{-1}$; on that event the chain-rule term is at most $S(X_i|T_i) \le \frac{1}{2}S(X_i) + C_{\text{buf}}$, and in every case it is at most $S(X_i)$; the probability of the event is only bounded below, so its positive error is charged in full. Taking the expectation of the chain rule,

$$\begin{aligned} F(Mr) &\le \Big[(1 - \tfrac{p_0}{2})M + \tfrac{p_0}{2}B\Big]F(r) + MC_{\text{buf}} \\ &\le \big(1 - \tfrac{p_0}{4}\big)M\,F(r) + CM, \end{aligned}$$

using $B \le M/2$. With $\lambda = (1 - p_0/4)M \in (1, M)$ for $r \ge w_0$, iterating at $r = w_0, Mw_0, M^2w_0, \dots$ gives $F(M^j w_0) \le C\lambda^j$ and hence $F(r) \le Cr^{e_0}$ with $e_0 = \log_M \lambda < 1$.

Part II
The one-dimensional matrix product state approximation

VII. STATEMENT, AND THE TWO-LINE ROUTE

In words. Theorem I.2 asks for one global vector, not for a bound on each cut. In two dimensions this is a serious gap between hypothesis and conclusion, and the source closes it with a distributed protocol. In one dimension there is a classical bridge: if the Schmidt spectrum at every cut has a fast-decaying tail, then truncating the cuts one after another produces a matrix product state whose error is the sum of the tail masses (Verstraete–Cirac). The marginal tail bound of Section III gives exactly such a tail, with an exponential rate u_0 that depends only on $q, R, J/\Delta$, and the area law fixes the location of the spectrum. So Theorem I.2 follows in a few lines, with bond dimension CL^{4/u_0}.

We record this first because it is true and short. The rest of Part II then presents the one-dimensional version of the *protocol* proof, which is the construction that generalizes to two dimensions and which uses none of this sequential truncation.

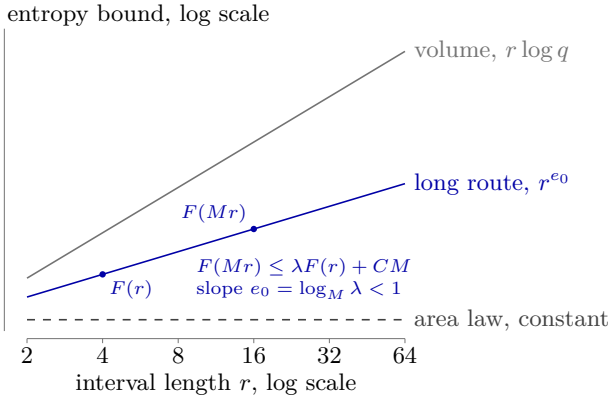


FIG. 9. Entropy bounds for an interval of length r on loglog axes (drawn with $M = 4$): the volume bound $r \log q$, parallel to the diameter r, has slope one; the long route gives slope $e_0 < 1$; the area law is the constant line reached directly in the previous stage.

Precisely.

Proposition VII.1 (Sequential truncation)**.** *Let Ω be a unit vector on $\mathcal{H}_\Lambda$, $\Lambda = \{1, \dots, L\}$. For $1 \le k \le L-1$ let Π_k be the projection of $\mathcal{H}_{[1,k]}$ onto the span of the D eigenvectors of $\rho_{[1,k]}$ with largest eigenvalues, let $\epsilon_k = \langle \Omega, (\mathrm{I} - \Pi_k \otimes \mathrm{I})\Omega \rangle$ be the discarded mass, and put*

$$\Phi = (\Pi_1 \otimes \mathrm{I})(\Pi_2 \otimes \mathrm{I}) \cdots (\Pi_{L-1} \otimes \mathrm{I})\,\Omega.$$

Then Φ is a matrix product state with open boundary conditions and bond dimension at most D, and $\|\Phi - \Omega\| \le \sum_k \sqrt{\epsilon_k}$.

Proof. Write $P_k = \Pi_k \otimes \mathrm{I}$. Telescoping, $\Phi - \Omega = \sum_{k=1}^{L-1} P_1 \cdots P_{k-1}(P_k - \mathrm{I})\Omega$ and each prefix is a contraction, which gives the error bound. For the bond dimension fix a cut k. The vector $P_k P_{k+1} \cdots P_{L-1}\Omega$ lies in $\operatorname{ran}\Pi_k \otimes \mathcal{H}_{[k+1,L]}$, so it has Schmidt rank at most D across the cut k, and the remaining factors $P_1, \dots, P_{k-1}$ act on $[1, k-1]$ and do not increase that rank. A vector with Schmidt rank at most D across every cut is a matrix product state of bond dimension at most D. □

Theorem VII.2 (Theorem I.2, two-line route)**.** *Under the hypotheses of Theorem I.2, for every $L \ge 2$ there is a matrix product state Φ with bond dimension at most*

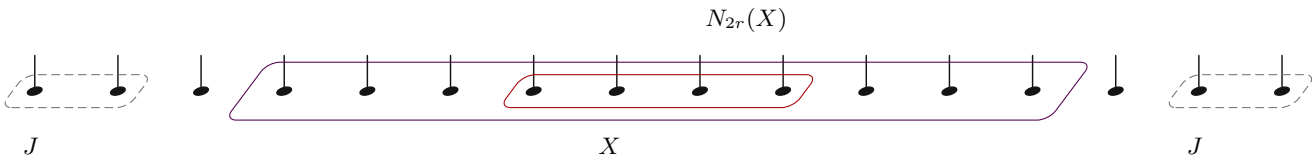


FIG. 10. Amplification on the chain: the interval X of length n, its graph neighbourhood $N_{2r}(X)$ with $r = O(n^{e_0/\alpha}) = o(n)$, on which the local unitary of the amplification argument acts, and the far set J it never touches, so that $I(X : J) \le n^{-M}$.

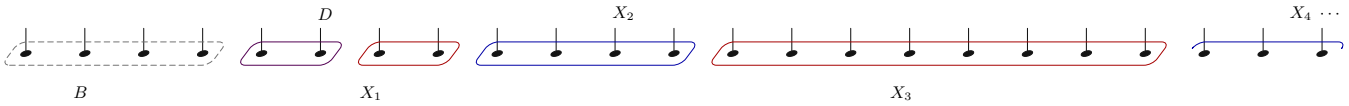


FIG. 11. Two families on a half-line: beyond the cut and the removed sites D, the dyadic intervals $X_i = [2^i, 2^{i+1})$ of lengths $2, 4, 8, \ldots$ with alternating family labels, continuing to the end of the chain. Each is separated from B and from all earlier same-family intervals by the previous interval, of half its length.

$2\,\mathrm{e}^{C_{\mathrm{int}}}\,(32\mathrm{e}^{\mathrm{e}/2}L^4)^{1/u_0}$ *and* $\min_\theta \|\Phi/\|\Phi\| - \mathrm{e}^{i\theta}\Omega\| \le L^{-1}$, *where* u_0 *is* (6) *and* C_{int} *is* (24).

Proof. Fix a cut k and $X = [1, k]$. By Lemma III.1, the eigenvalues p of ρ_X with $-\log p > S(X) + w$ have total mass at most $2\mathrm{e}^{\mathrm{e}/2}\mathrm{e}^{-u_0 w}$; the eigenvalues with $-\log p \le S(X) + w$ number at most $\mathrm{e}^{S(X)+w}$. Choose w with $2\mathrm{e}^{\mathrm{e}/2}\mathrm{e}^{-u_0 w} = (16L^4)^{-1}$ and $D = \lceil \mathrm{e}^{C_{\mathrm{int}}+w} \rceil \ge \mathrm{e}^{S(X)+w}$ (Theorem I.1). Then $\epsilon_k \le (16L^4)^{-1}$ for every k, Proposition VII.1 gives $\|\Phi - \Omega\| \le (L-1)/(4L^2) \le 1/(4L)$, hence $\Phi \ne 0$ and $\|\Phi/\|\Phi\| - \Omega\| \le 2\|\Phi - \Omega\| \le L^{-1}$. The bond dimension is $D \le 2\mathrm{e}^{C_{\mathrm{int}}+w}$ with $\mathrm{e}^w = (32\mathrm{e}^{\mathrm{e}/2}L^4)^{1/u_0}$. □

Remark VII.3. The exponent $4/u_0$ is of order $\sqrt{(1 + J/\Delta)\mathcal{B}_R}$. Hastings' original construction gives a polynomial bond dimension with an exponent that is exponential in J/Δ; here the exponent is a square root. This is a byproduct of the tail bound, not of the protocol below.

Plan of the protocol route

In words. The protocol proof has four layers, and we keep all four.

1. *Information at logarithmic separation.* Two regions of the chain separated by $D \log L$ sites share at most L^{-p} nats of mutual information. This is where the gap is used. In two dimensions it took the coarse collar bound, the cell conditional-information lemma and the "angular" induction; here it takes the area law plus one lemma, the information reset (Section VIII).
2. *Moves.* Three elementary operations change who holds which registers while keeping the global vector close to a reference: a birth (give an interior interval to a new owner), its reverse, a death, and an exchange of assignments between two copies of the ground state inside a window. Each is a contraction built from private maps and one normalized bipartite vector or covector, and each costs $O(L^{-p/2})$ when the moved region is informationally independent of the other owners (Section IX).
3. *The schedule.* A left-to-right sweep with a moving remainder party creates the block owners one by one. The only difficulty is a band owned by the remainder that is wedged between two different owners; it is dissolved by a fixed sequence of nine moves using an auxiliary copy of the ground state, copied from the edge construction of the two-dimensional paper. Every party takes part in a bounded number of moves (Section X).
4. *Compression and assembly.* The distributed construction is a pure linear calculation with unbounded private dimensions. Replacing each bipartite resource by an average of polynomially many random product operators approximates the physical density in trace norm, and the error analysis is dimension-free because each party participates in boundedly many operations. One column of the resulting operator network is a ket network; placing its nodes on the chain gives the matrix product state (Sections XI, XII).

VIII. INFORMATION AT LOGARITHMIC SEPARATION

A. Purifications, fidelity, and splitting

In words. Small mutual information $I(T : E)$ means the joint state of T and E is close, in relative entropy, to the product of its marginals. Relative entropy controls fidelity, and fidelity between two mixed states is the largest overlap between their purifications (Uhlmann). So if the third system U purifies TE, there is an isometry on U alone that turns the global vector into nearly a product of a purification of T and a purification of E. Whoever holds U can do this privately. This is the single mechanism behind births and exchanges.

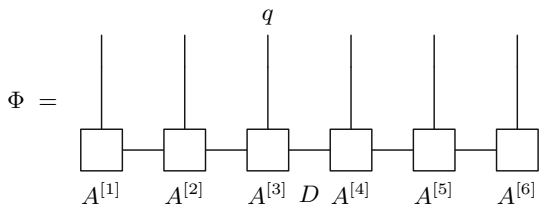


FIG. 12. A matrix product state with open boundary conditions: one tensor per site, bonds of dimension D between neighbours, one physical leg per site.

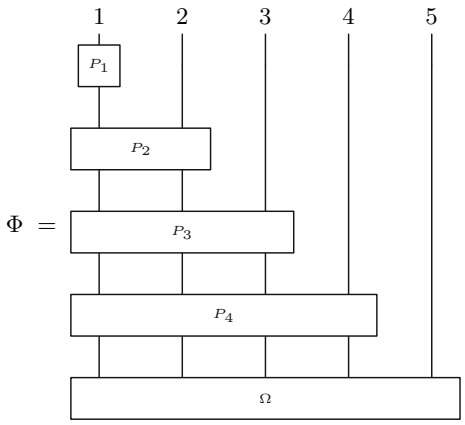


FIG. 13. Sequential truncation, drawn for $L = 5$. The projector $P_k = \Pi_k \otimes 1$ onto the top D Schmidt vectors of the prefix $[1, k]$ is applied for $k = L - 1$ first and $k = 1$ last; the output of P_k lies in $\operatorname{ran}\Pi_k \otimes \mathcal{H}_{[k+1,L]}$, so across every cut the result has Schmidt rank at most D.

Precisely. For density operators ρ, σ put $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ and $D(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$, infinite when the support of ρ is not contained in that of σ.

Lemma VIII.1 (Fidelity, information, Uhlmann)**.** *For normalized finite-dimensional states, $F(\rho, \sigma) \geq \mathrm{e}^{-D(\rho\|\sigma)/2}$, and $D(\rho_{AC}\|\rho_A \otimes \rho_C) = I(A : C)$. If ξ purifies ρ on $\mathcal{H} \otimes \mathcal{K}$ and η purifies σ on $\mathcal{H} \otimes \mathcal{K}'$, then after enlarging $\mathcal{K}'$ if necessary there is an isometry $V : \mathcal{K} \to \mathcal{K}'$ with $|\langle\eta, (\mathrm{I} \otimes V)\xi\rangle| = F(\rho, \sigma)$.*

Proof. Let $(p_i, |i\rangle)$, $(q_j, |j'\rangle)$ be eigenpairs of ρ, σ. If $D(\rho\|\sigma) < \infty$, $w_{ij} = p_i|\langle i|j'\rangle|^2$ is a probability on the pairs with $p_i q_j > 0$ and $D(\rho\|\sigma) = \sum w_{ij} \log(p_i/q_j)$. By convexity of the exponential, $\operatorname{Tr}\sqrt{\rho}\sqrt{\sigma} = \sum_{ij} w_{ij} \exp(-\frac{1}{2}\log(p_i/q_j)) \geq \mathrm{e}^{-D(\rho\|\sigma)/2}$, and the trace norm dominates the trace. The identity for $I(A : C)$ is the expansion of $\log(\rho_A \otimes \rho_C)$. The purification statement is Uhlmann's theorem: the overlaps $\operatorname{Tr}(\sqrt{\rho}\sqrt{\sigma}U)$ over unitaries U on a common large purifying space have maximum modulus $\|\sqrt{\rho}\sqrt{\sigma}\|_1$ by polar decomposition, and any purification is obtained from the canonical one by an isometry on the purifying factor. □

Lemma VIII.2 (Splitting)**.** *Let T, U, E partition Λ and $b = I_\Omega(T : E)$. There are finite-dimensional spaces $\mathcal{B}_T, \mathcal{B}_E$, an isometry $V : \mathcal{H}_U \to \mathcal{B}_T \otimes \mathcal{B}_E$ and unit vectors $s \in \mathcal{H}_T \otimes \mathcal{B}_T$, $s' \in \mathcal{H}_E \otimes \mathcal{B}_E$ with*

$$\|(\mathrm{I}_{TE} \otimes V)\Omega - s \otimes s'\| \leq \sqrt{2(1 - \mathrm{e}^{-b/2})} \leq \sqrt{b}.$$

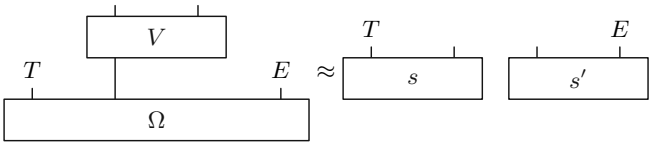


FIG. 14. The splitting lemma as a diagram: the isometry V on the buffer U turns Ω into nearly a product of a purification s of ρ_T and a purification s' of ρ_E.

Proof. ρ_{TE} is supported in $\operatorname{supp}\rho_T \otimes \operatorname{supp}\rho_E$ (the projector onto the kernel of a marginal, tensored with the identity, has zero expectation in the positive joint state), so $D(\rho_{TE}\|\rho_T \otimes \rho_E) = b$ is finite. Purify ρ_T by s and ρ_E by s', enlarge the purifying factors so that $\mathcal{B}_T \otimes \mathcal{B}_E$ contains an isometric copy of $\mathcal{H}_U$, and apply Lemma VIII.1: an isometry V on $\mathcal{H}_U$ attains overlap at least $\mathrm{e}^{-b/2}$ with $s \otimes s'$, with real positive phase. Two unit vectors with overlap F are at distance $\sqrt{2(1 - F)}$. □

B. The information reset

In words. The area law gives $I(R : C) = O(1)$ between an interval R and anything else, never $o(1)$. The reset lemma turns a constant into an arbitrary inverse power of L by shrinking the two regions inward by a margin of order

$\log L$. The idea is a two-replica swap. Take two copies of the ground vector, $\Psi = \Omega \otimes \Omega$, and swap the two copies on R to get $\Phi = F_R\Psi$. In Φ, the core $R_0 \subseteq R$ of one copy and the core $C_0 \subseteq C$ of the same copy come from different original copies, so they are exactly independent. If we can find a unitary U acting only on the buffer between the cores and sending Ψ to Φ up to L^{-p}, then the joint marginal of the cores in Ψ equals the product marginal up to L^{-p}, which is the claim.

The unitary is built in three steps. First, $I(R:C) \le b$ and the splitting lemma give an operator W on the buffer Q of both copies with $\langle \Phi, W\Psi\rangle \ge \mathrm{e}^{-2b}$. Second, a Gaussian spectral filter applied in both directions, $M = \int g_h(t)\mathrm{e}^{it\mathbb{H}'}W\mathrm{e}^{-it\mathbb{H}}dt$, projects $W\Psi$ onto Φ and $W^\dagger\Phi$ onto Ψ with error $\mathrm{e}^{-h\Delta^2/2}$, because Ψ, Φ are the unique gapped ground vectors of the two doubled Hamiltonians $\mathbb{H}$, $\mathbb{H}' = F_R\mathbb{H}F_R$. Third, a Lieb–Robinson bound localizes M to within $\ell = O(1+b+\log L)$ of the buffer and of the edges crossing R, where $\mathbb{H}$ and $\mathbb{H}'$ differ. The localized operator is almost isometric on Ψ, and its polar unitary does the job. The lemma is proved exactly as in the source; we only insert the one-dimensional locality bound.

Precisely. We need the propagation estimate for a doubled chain. Let $\mathbb{H}$ be a Hamiltonian on $\bigotimes_{x\in\Lambda}(\mathbb{C}^q \otimes \mathbb{C}^q)$ which is a sum of terms k_Z supported on sets Z of diameter at most R with $\sum_{Z\ni x}\|k_Z\| \le b_0$ for every x, and $\tau_t(A) = \mathrm{e}^{it\mathbb{H}}A\mathrm{e}^{-it\mathbb{H}}$.

Lemma VIII.3 (Lieb–Robinson bound on a chain)**.** *There are $C, v, \mu > 0$ depending only on R and b_0 such that for every operator A supported in $S \subseteq \Lambda$, every site y and every operator B_y supported at y,*

$$\|[\tau_t(A), B_y]\| \le C|S|\,\|A\|\,\|B_y\|\,\mathrm{e}^{v|t|-\mu\,\mathrm{dist}(S,y)}.$$

Consequently, if $\mathcal{P}_D$ denotes the conditional expectation onto operators supported in D (the average over independent Haar unitaries at the sites outside D), then for every $\ell \ge 0$

$$\|\tau_t(A) - \mathcal{P}_{N_\ell(S)}\tau_t(A)\| \le C'|S|^2\,\|A\|\,\mathrm{e}^{v|t|-\mu\ell}.$$

Proof. For $t \ge 0$ put $c_y(S,t) = \sup\{\|[\tau_t(A), B_y]\| : \operatorname{supp} A \subseteq S, \|A\| \le 1\}$. Differentiating and using the Jacobi identity, the derivative of $[\tau_t(A), B_y]$ equals $i[\sum_{Z\cap S\neq\varnothing}\tau_t(k_Z), [\tau_t(A), B_y]]$ plus a sum over $Z \cap S \neq \varnothing$ of $-i[\tau_t(A), [\tau_t(k_Z), B_y]]$; the first part is removed by a unitary change of frame, so $c_y(S,t) \le c_y(S,0) + 2\sum_{Z\cap S\neq\varnothing}\|k_Z\|\int_0^t c_y(Z,u)\,du$. Iterating, and using $c_y(Z,u) \le 2\|B_y\|$ to see that the remainder after m iterations tends to zero, gives

$$c_y(S,t) \le 2\|B_y\| \sum_{m\ge 0}\frac{(2t)^m}{m!}\sum_{\substack{Z_1\cap S\neq\varnothing,\ Z_{i+1}\cap Z_i\neq\varnothing \\ y\in Z_m}}\prod_{i=1}^{m}\|k_{Z_i}\|,$$

with the $m = 0$ term equal to $\mathbf{1}_{\{y\in S\}}$. Summing the norms of the terms meeting a set of n sites gives at most nb_0, so the inner sum is at most $|S|\,b_0\,((R+1)b_0)^{m-1} \le |S|\kappa^m$ with $\kappa = (R+1)b_0$; and a chain reaching y needs $mR \ge \mathrm{dist}(S,y)$. Hence for every $\mu > 0$, $c_y(S,t) \le 2|S|\|B_y\|\mathrm{e}^{-\mu\,\mathrm{dist}(S,y)}\sum_m(2\kappa\mathrm{e}^{\mu R}t)^m/m!$, which is the first bound with $v = 2\kappa\mathrm{e}^{\mu R}$. Negative times are identical. For the second bound, $\mathcal{P}_D$ is a product of commuting single-site twirls T_y, telescoping gives $\|B - \mathcal{P}_D B\| \le \sum_{y\notin D}\sup_{U_y}\|[B, U_y]\|$, and there are at most $2|S|$ sites at each distance $d > \ell$ from S, so the sum is at most $4C|S|^2\|A\|\mathrm{e}^{vt}\sum_{d>\ell}\mathrm{e}^{-\mu d}$. □

Lemma VIII.4 (Information reset)**.** *For every $p > 0$ there is $C_p < \infty$, depending only on p, q, R, J, Δ, with the following property for all sufficiently large L. Let $R, C \subseteq \Lambda$ be disjoint, $R_0 \subseteq R$, $C_0 \subseteq C$, and suppose $I_\Omega(R:C) \le b$. If*

$$\mathrm{dist}(R_0, \Lambda\setminus R) > w, \qquad \mathrm{dist}(C_0, \Lambda\setminus C) > w,$$
$$w = C_p(1 + b + \log L),$$

then $I_\Omega(R_0 : C_0) \le L^{-p}$.

Proof. If a core is empty there is nothing to prove. Let $\Psi = \Omega\otimes\Omega$ on two replicas, F_R the unitary exchanging the two replicas at the sites of R, $\Phi = F_R\Psi$, and $Q = \Lambda\setminus(R\cup C)$.

Overlap. Apply Lemma VIII.2 to $(T, U, E) = (R, Q, C)$: an isometry $V : \mathcal{H}_Q \to \mathcal{B}_R \otimes \mathcal{B}_C$ and unit vectors $a \in \mathcal{H}_R\otimes\mathcal{B}_R$, $a' \in \mathcal{H}_C\otimes\mathcal{B}_C$ with $|\langle a\otimes a', \widetilde{\Omega}\rangle| \ge \mathrm{e}^{-b/2}$, $\widetilde{\Omega} = (\mathrm{I}\otimes V)\Omega$. The largest Schmidt probability (eigenvalue of the marginal) of $\widetilde{\Omega}$ across $R\mathcal{B}_R\,|\,C\mathcal{B}_C$ is therefore at least e^{-b}, so $\mathrm{Tr}\,\rho^2_{\widetilde{\Omega}, R\mathcal{B}_R} \ge \mathrm{e}^{-2b}$. Define on the two replicas of Q the Hermitian contraction $W = (V^\dagger\otimes V^\dagger)F_{\mathcal{B}_R}(V\otimes V)$, where $F_{\mathcal{B}_R}$ swaps the two copies of $\mathcal{B}_R$. Since V acts outside R, the swap identity $\langle\xi^{\otimes 2}, F_A\xi^{\otimes 2}\rangle = \mathrm{Tr}\,\rho^2_{\xi,A}$ gives

$$\begin{aligned} z := \langle\Phi, W\Psi\rangle &= \langle\widetilde{\Omega}^{\otimes 2}, F_R F_{\mathcal{B}_R}\widetilde{\Omega}^{\otimes 2}\rangle \\ &= \mathrm{Tr}\,\rho^2_{\widetilde{\Omega}, R\mathcal{B}_R} \ \ge\ \mathrm{e}^{-2b}. \end{aligned} \tag{25}$$

Two-sided Gaussian filtering. Let $\mathbb{H} = H\otimes\mathrm{I} + \mathrm{I}\otimes H$ and $\mathbb{H}' = F_R\mathbb{H}F_R$. Both are range-$R$ Hamiltonians on the doubled chain with uniformly bounded interaction sums, with unique ground vectors Ψ, Φ, common ground energy $\mathcal{E} = 2E_0$, and gap at least Δ. Their difference $\mathbb{H}' - \mathbb{H}$ is a sum of at most $N_R|\partial R|$ terms of norm at most $2J$, those of $\mathbb{H}$ whose support meets both R and $\Lambda\setminus R$ (interior terms of R are merely exchanged between the two identical replica copies); let Z_R be the union of their supports, a set within distance R of the cut edges of R. Put $B_* = 1 + b + \log L$, $r_* = 2p + 10$, $h = AB_*$ with $A\Delta^2/2 \ge r_* + 30$, and

$$g_h(t) = (2\pi h)^{-1/2}\mathrm{e}^{-t^2/(2h)},$$
$$M = \int_{\mathbb{R}} g_h(t)\,\mathrm{e}^{it\mathbb{H}'}W\mathrm{e}^{-it\mathbb{H}}\,dt.$$

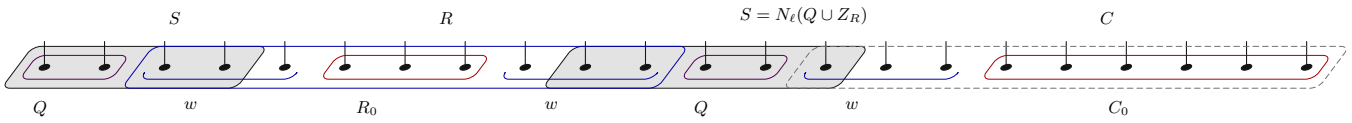


FIG. 15. The reset geometry, drawn in one replica of $\Psi = \Omega \otimes \Omega$: the cores $R_0 \subset R$ and $C_0 \subset C$, the buffer $Q = \Lambda \setminus (R \cup C)$, the margins of width w, the set Z_R near the cut edges of R, and the support $S = N_\ell(Q \cup Z_R)$ of the unitary, which stays clear of both cores.

M is a contraction, and by the spectral theorem $M\Psi = \mathrm{e}^{-h(\mathbb{H}'-\mathcal{E})^2/2} W\Psi$, $M^\dagger\Phi = \mathrm{e}^{-h(\mathbb{H}-\mathcal{E})^2/2} W^\dagger\Phi$, so by the gaps

$$\|M\Psi - z\Phi\|,\ \|M^\dagger\Phi - z\Psi\| \le \mathrm{e}^{-h\Delta^2/2} \le \mathrm{e}^{-10b} L^{-r_*}. \quad (26)$$

Truncating the integral to $|t| \le T = T_0 B_*$ costs at most $2\mathrm{e}^{-T_0^2 B_*/(2A)}$, which is also at most $\mathrm{e}^{-10b} L^{-r_*}$ once $T_0^2/(2A) \ge r_* + 30$.

Localization. Write $\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}} = (\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}'})\, V(t)$ with $V(t) = \mathrm{e}^{it\mathbb{H}'}\mathrm{e}^{-it\mathbb{H}}$, whose left generator $G(t) = \mathrm{e}^{it\mathbb{H}'}(\mathbb{H}'-\mathbb{H})\mathrm{e}^{-it\mathbb{H}'}$ is a sum of the boundedly many terms supported in Z_R. Lemma VIII.3 applied to W (support Q, norm one) and to the terms of $G(t)$ gives Hermitian localized approximants $W_\ell(t) = \mathcal{P}_{N_\ell(Q)}\big(\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}'}\big)$ and $G_\ell(t) = \mathcal{P}_{N_\ell(Z_R)} G(t)$ with

$$\|\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}'} - W_\ell(t)\| + \|G(t) - G_\ell(t)\| \le C L^2\, \mathrm{e}^{v|t|-\mu\ell},$$

where the factor L^2 bounds $|Q|^2$ and $|\partial R|$. Let $V_\ell(t)$ be the unitary evolution generated by $G_\ell(t)$, supported in $N_\ell(Z_R)$; Duhamel's formula gives $\|V(t) - V_\ell(t)\| \le CL^2|t|\mathrm{e}^{v|t|-\mu\ell}$. Hence $W_\ell(t)V_\ell(t)$ is a contraction supported in $S := N_\ell(Q \cup Z_R)$ which approximates the integrand uniformly for $|t| \le T$ with error $CL^2(1+T)\mathrm{e}^{vT-\mu\ell}$. Take $\ell = \lceil K_{\mathrm{loc}} B_* \rceil$ with K_{loc} fixed large enough (after T_0) that this error is at most $\mathrm{e}^{-10b} L^{-r_*}$, using $\log L \le B_*$ and $\log(1+T) \le C + B_*$. Averaging the localized integrands against the truncated Gaussian gives a contraction M_{loc} supported in S with

$$\|M_{\mathrm{loc}}\Psi - z\Phi\|,\ \|M_{\mathrm{loc}}^\dagger\Phi - z\Psi\| \le C\mathrm{e}^{-10b} L^{-r_*}. \quad (27)$$

Polar decomposition. Put $D = M_{\mathrm{loc}}/z$. Then $\|D\| \le \mathrm{e}^{2b}$, $\|D\Psi - \Phi\|, \|D^\dagger\Phi - \Psi\| \le C\mathrm{e}^{-8b} L^{-r_*}$, and writing $(D^\dagger D - \mathrm{I})\Psi = D^\dagger(D\Psi - \Phi) + (D^\dagger\Phi - \Psi)$, $\|(D^\dagger D - \mathrm{I})\Psi\| \le C\mathrm{e}^{-6b} L^{-r_*}$. The polar partial isometry of the square operator D on $\mathcal{H}_S$ extends to a unitary U on $\mathcal{H}_S$ (its two support spaces have equal dimension), tensored with the identity elsewhere. Since $(\sqrt{x}-1)^2 \le (x-1)^2$ for $x \ge 0$, $\|(U-D)\Psi\| = \|(\mathrm{I} - |D|)\Psi\| \le \|(\mathrm{I} - D^\dagger D)\Psi\|$, so

$$\|U\Psi - \Phi\| \le C\mathrm{e}^{-6b} L^{-r_*} \le CL^{-r_*}. \quad (28)$$

Support and conclusion. Every site of Q is more than w from R_0 and from C_0. A term whose support meets both R and $\Lambda \setminus R$ has a site $y \notin R$, which is more than w from R_0, and a site $x \in R \subseteq \Lambda \setminus C$, which is more than w from C_0; its support has diameter at most R, so every site of Z_R is more than $w - R$ from both cores. Choosing $C_p > K_{\mathrm{loc}} + R + 3$ makes $S = N_\ell(Q \cup Z_R)$ disjoint from both cores in both replicas. Thus U does not change the joint two-replica marginal on R_0C_0. In $\Phi = F_R\Psi$, the marginal on R_0 and C_0 of the *first* replica is $\rho_{R_0} \otimes \rho_{C_0}$, because R_0 was swapped in from the second replica and C_0 was not. By (28) and contractivity of the partial trace, the original marginal $\rho_{R_0C_0}$ is within trace distance CL^{-r_*} of $\rho_{R_0} \otimes \rho_{C_0}$. The continuity bound $|S(A|B)_\rho - S(A|B)_\sigma| \le C\sqrt{\delta}\log(\mathrm{e}\dim\mathcal{H}_A)$ for $\delta = \frac{1}{2}\|\rho - \sigma\|_1$ [6], applied to $I(R_0 : C_0) = S(R_0) - S(R_0|C_0)$ with $\log\dim\mathcal{H}_{R_0} \le L\log q$, gives $I_\Omega(R_0 : C_0) \le CL^{1-r_*/2} = CL^{-p-4} \le L^{-p}$ for large L. □

C. The one-dimensional separation lemma

In words. Combine the two previous facts. An interval R has entropy at most C_{int}, so its mutual information with anything is at most $2C_{\mathrm{int}}$. Feed this constant into the reset lemma: any interval T and any set E at distance of order $\log L$ from T are independent up to L^{-p}. This replaces the entire Section 3 of the two-dimensional paper (coarse collar, cell conditional information, angular induction), and it holds for intervals of every length, from a few sites to the whole chain.

Precisely.

Lemma VIII.5 (Information at logarithmic separation)**.** *For every $p > 0$ and $k_0 \ge 1$ there is $D_p = D_p(q, R, J, \Delta, k_0)$ such that for all sufficiently large L, every $T \subseteq \Lambda$ that is a union of at most k_0 intervals, and every $E \subseteq \Lambda$ with $\mathrm{dist}(T, E) \ge D_p \log L$,*

$$I_\Omega(T : E) \le L^{-p}.$$

Proof. Let $b = 4k_0C_{\mathrm{int}}$, $w = C_p(1 + b + \log L)$ from Lemma VIII.4, and $D_p \log L \ge 3w+3$. Put $R = T^{+(w+1)}$, a union of at most k_0 intervals, $C = \Lambda \setminus T^{+(2w+2)}$, $R_0 = T$, $C_0 = E$. Then $\mathrm{dist}(R_0, \Lambda \setminus R) \ge w + 1$, $\mathrm{dist}(C_0, \Lambda \setminus C) \ge w + 1$, and $I(R : C) \le 2S(R) \le 2k_0C_{\mathrm{int}} \le b$ by Theorem I.1 (subadditivity over the at most k_0 intervals of R, and $I(R : C) = S(R) + S(C) - S(RC) \le 2S(R)$ by the Araki–Lieb inequality). Lemma VIII.4 gives the claim. □

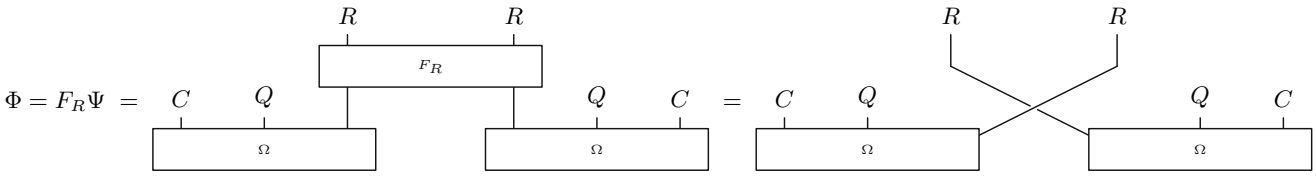


FIG. 16. The swap F_R exchanges the two copies of Ω on R only (the first copy is drawn mirrored so that the two R legs are adjacent). In $\Phi = F_R\Psi$ the core R_0 of the first copy comes from the second copy while C_0 does not, so the first copy's marginal on R_0C_0 is $\rho_{R_0} \otimes \rho_{C_0}$.

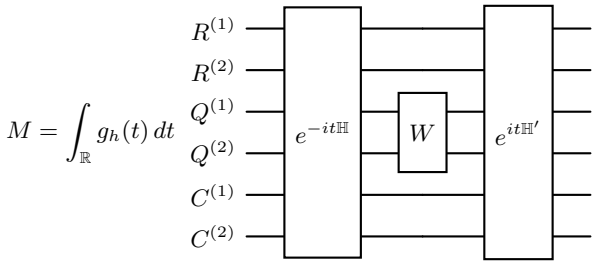


FIG. 17. The two-sided Gaussian filter: evolve with the doubled Hamiltonian, apply W on the two copies of the buffer, evolve back with the swapped Hamiltonian, and average against a Gaussian in time. It sends $W\Psi$ to a multiple of Φ and $W^\dagger\Phi$ to a multiple of Ψ, up to $e^{-h\Delta^2/2}$.

Remark VIII.6 (A route without Lieb–Robinson bounds)**.** The reset lemma is the only place in Part II where a Lieb–Robinson bound is used. It can be replaced by the amplification argument of [1, Section 10], which uses exact positive quasi-local constraints with stretched-exponential tails e^{-cl^α}, $\alpha = p/(p{+}1)$, random products of the associated contractions, and a polar decomposition, and needs no propagation bound for the unitary dynamics. With the area law as input, that argument gives $I(T : E) \le L^{-M}$ for E beyond distance $C(\log L)^{1/\alpha}$ of an interval T. Running the protocol below with blocks of that size yields a matrix product state with bond dimension $\exp(O((\log L)^{1/\alpha}))$, quasi-polynomial for every fixed p, and polynomial only in the limit. Exactly polynomial bond dimension through the protocol requires the logarithmic separation of Lemma VIII.5.

IX. DISTRIBUTED CONSTRUCTIONS AND THE THREE MOVES

In words. Think of a bookkeeping game. There are *parties*; each owns a set of registers (tensor factors), which may include whole copies of the ground vector held privately. A *gate* is a linear contraction applied to the registers of a few parties. A gate is *private* if it touches one party only; private gates are free. A non-private gate is charged only through two numbers: how many parties it involves, and how many such gates each party ever takes part in. In addition every non-private gate must have a *polynomial expansion*: it must be a sum of polynomially many terms, each a composition of private maps, preparations of normalized vectors shared by two parties, contractions by normalized covectors shared by two parties, and transfers of polynomial-dimensional registers. The dimensions of the shared vectors are unrestricted.

The point of the game is that its final physical density admits a tensor network with polynomial links on the party graph (Section XI), whatever the private dimensions were. So we are free to prepare the ground vector exactly at one party and to use arbitrarily entangled bipartite resources; the price is only paid in participation counts.

A *sheet* is a full set of site registers with an assignment of owners; its reference vector is a copy of the ground vector placed on those registers in canonical site coordinates (ownership does not permute sites). The three moves change the ownership of a sheet or of two sheets while keeping the actual vector within $O(L^{-p/2})$ of the product of the reference vectors.

Precisely. A *distributed construction* of size parameter L consists of a finite set of parties, each with an initial finite-dimensional register space in a normalized product initial vector, and a sequence of gates, each a linear contraction between specified input and output register spaces of the participating parties, the other registers being spectators. Registers to be discarded are retained until a final partial trace. An *allowed monomial* is a composition of private contractions, preparations of normalized vectors shared between two participating parties, contractions by normalized covectors shared between two participating parties, and identity transfers of registers of uniformly polynomial dimension. A non-private gate is *admissible* if it is a sum of uniformly polynomially many allowed monomials with uniformly polynomially bounded scalar coefficients, each monomial containing a bounded number of shared vectors, shared covectors and transfers; a contraction within L^{-a} of such a sum, for a fixed $a > 0$,

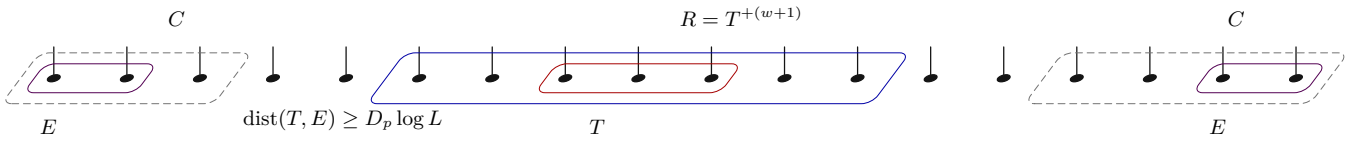


FIG. 18. The separation lemma: an interval T, its enlargement R, the far region C and the set $E \subseteq C$ at distance $D_p \log L$ from T. The area law bounds $I(R:C)$ by a constant and the reset makes $I(T:E) \le L^{-p}$.

is also admissible after rescaling by $1 + L^{-a}$. A *bounded change* is a sequence of a bounded number of admissible gates on a bounded number of parties.

A sheet is $\mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} \mathbb{C}^q$ with an owner assigned to each site; its reference vector is Ω in these registers.

Lemma IX.1 (Birth and death)**.** *In a sheet, let T be a set of sites owned by $P_\circ$, and let $E = \{x :$ the owner of x is not $P_\circ\}$, $U = \Lambda \setminus (T \cup E)$. If $I_\Omega(T : E) \le L^{-2p}$, then the change of owner of T from $P_\circ$ to any party $Q_\circ$ (a* birth*) is a bounded change involving $P_\circ, Q_\circ$ only, consisting of private maps and one shared normalized vector, with reference-vector error at most L^{-p}. The reverse change (a* death*: T returns from $Q_\circ$ to $P_\circ$) is a bounded change with the same error whenever the ownership and information conditions hold in the sheet* after *the death; it uses one shared normalized covector instead.*

Proof. Let V, s, s' be given by Lemma VIII.2 for (T, U, E), put $\mathcal{V} = \mathrm{I}_{TE} \otimes V$ and $\Pi = |s\rangle\langle s|_{T\mathcal{B}_T} \otimes \mathrm{I}_{E\mathcal{B}_E}$. All of T, U belong to $P_\circ$. At $P_\circ$ apply V on U; contract $T\mathcal{B}_T$ with $\langle s|$ (private, both at $P_\circ$); supply the shared vector s on $T\mathcal{B}_T$ with T now owned by $Q_\circ$ and $\mathcal{B}_T$ by $P_\circ$; apply $V^\dagger$ at $P_\circ$. The composite is the Hermitian contraction $B = \mathcal{V}^\dagger \Pi \mathcal{V}$ and $\|(B - \mathrm{I})\Omega\| \le \|(\Pi - \mathrm{I})\mathcal{V}\Omega\| = \|(\Pi - \mathrm{I})(\mathcal{V}\Omega - s \otimes s')\| \le L^{-p}$. For the death, apply V at $P_\circ$, use the shared covector $\langle s|$ on $T\mathcal{B}_T$ held by $Q_\circ, P_\circ$, reinsert s wholly at $P_\circ$, and apply $V^\dagger$: the same operator B with reversed ownership. □

Lemma IX.2 (Two-sheet exchange)**.** *Consider two sheets and an interval $Y \subseteq \Lambda$. Inside Y exchange the two ownership assignments (sheet 1 at $x \in Y$ gets the owner sheet 2 had, and conversely); outside Y keep both. Let $P_\circ$ be a party,*

$$Z = \{x : \text{the two owners of } x \text{ are not both } P_\circ\},$$
$$T = Z \cap Y, \quad E = Z \setminus Y, \quad U = \Lambda \setminus Z.$$

If $I_\Omega(T : E) \le L^{-2p}$, the exchange is implemented by renaming of registers (no register changes party) and private contractions at $P_\circ$, with error at most $4L^{-p}$ between the products of the old and of the new reference vectors.

Proof. Let $\mathcal{R}$ rename the two sheets inside Y: a private identification of tensor factors, since the new owner of sheet 1 at $x \in Y$ is the old owner of sheet 2 there. In canonical coordinates $\mathcal{R}$ acts as the swap F_Y of the two sheets on Y. All registers of U on both sheets are held by $P_\circ$ before and after. Let V, s, s' be given by Lemma VIII.2 for (T, U, E), $\mathcal{V} = \mathrm{I}_{TE} \otimes V$, and let $D_U = (V^{\otimes 2})^\dagger F_{\mathcal{B}_T} V^{\otimes 2}$ on the two copies of U, a private contraction at $P_\circ$ (apply V to both copies, swap the two $\mathcal{B}_T$ factors, apply the adjoints). With $A = Y \cap U$, after $\mathcal{R}$ apply the private swap F_A and then D_U. Since $Y = T \sqcup A$, the net action on the two copies is

$$D_U F_A F_Y = D_U F_T = (\mathcal{V}^{\otimes 2})^\dagger F_{T\mathcal{B}_T} \mathcal{V}^{\otimes 2},$$

using that F_T acts outside U and commutes through $V^{\otimes 2}$. Put $\zeta = s \otimes s'$ and $\delta = \|\mathcal{V}\Omega - \zeta\| \le L^{-p}$. Then $\|(\mathcal{V}\Omega)^{\otimes 2} - \zeta^{\otimes 2}\| \le 2\delta$, $\zeta^{\otimes 2}$ is fixed by $F_{T\mathcal{B}_T}$ (it swaps two identical factors s), and $(\mathcal{V}^{\otimes 2})^\dagger \mathcal{V}^{\otimes 2} = \mathrm{I}$, so

$$\begin{aligned}&\big\|(\mathcal{V}^{\otimes 2})^\dagger F_{T\mathcal{B}_T} \mathcal{V}^{\otimes 2} \Omega^{\otimes 2} - \Omega^{\otimes 2}\big\| \\ &\qquad \le \|(F_{T\mathcal{B}_T} - \mathrm{I})\big((\mathcal{V}\Omega)^{\otimes 2} - \zeta^{\otimes 2}\big)\| \le 4\delta.\end{aligned}$$

In canonical coordinates both the old and the new reference vector are $\Omega \otimes \Omega$ (the sheets are relabeled, not moved), which is the claim. □

Remark IX.3. In the two-dimensional source these lemmas carry encoded holes and tags at the points where three owners meet. On a line, the closure of two intervals with different owners meets at most two owners at any point, so there are no such points and no encodings. This is the first of the two reasons why the adaptive patch constraints of [2, Section 4] are not needed in one dimension.

X. THE SWEEP PROTOCOL

In words. Here is the schedule. Divide the chain into consecutive blocks of t sites, where $t = 20w$ and w is the logarithmic separation of Lemma VIII.5; the last block is whatever is left over, of length between $3t$ and $4t$. At the start, a single party R_0 holds everything. At step k the picture is

$$P_1 \,|\, P_2 \,|\, \cdots \,|\, P_k \,|\, R_k,$$

with P_i owning block i and the *remainder* party R_k owning the rest of the chain. Two births, both allowed because they leave a margin of w to any other owner, create the next remainder R_{k+1} on everything beyond block $k+1$ plus a margin, and the next block owner P_{k+1} on the interior of block $k+1$. What remains of R_k are two bands of width w: one wedged between P_k and P_{k+1}, one between P_{k+1} and R_{k+1}. Each band must be dissolved, and this is the only nontrivial step.

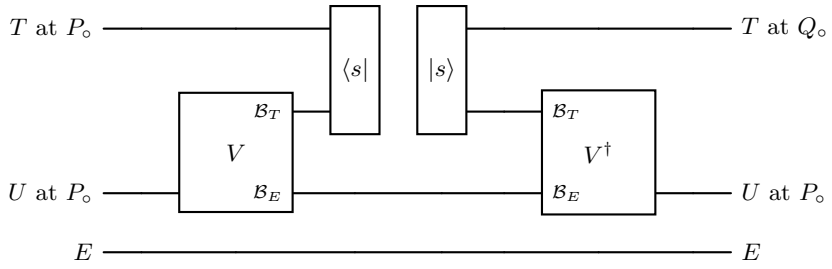


FIG. 19. The birth circuit, read left to right: V splits the buffer, $\langle s|$ closes T and its purifier, a fresh pair s re-emits them with T held by the new owner, and $V^\dagger$ recombines.

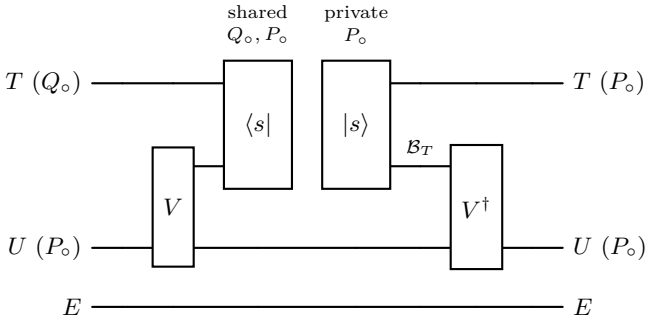


FIG. 20. The death circuit: the same operator with the ownership reversed, using the shared covector $\langle s|$ across the two owners and a private re-insertion of s.

A band A wedged between two different owners C (left) and B (right) cannot simply die into B: the band touches C, so it is not informationally independent of the non-B owners. The two-dimensional paper solves this along every edge of every square with a fixed sequence of moves using an auxiliary copy of the ground vector, and the sequence transplants verbatim. In units of the band width, with the band at $0 < x < 1$, C to the left and B to the right:

1. Create a fresh copy of the ground vector, all of it owned by C. On it, birth B on $(-6, -1)$, then A on $(-5, -2)$, then C on $(-4, -3)$. The auxiliary word is $C\,|\,B\,|\,A\,|\,C\,|\,A\,|\,B\,|\,C$.
2. Exchange the two sheets inside the window $(-7, -\frac{7}{2})$. Both ends of the window are in places where both sheets are owned by C, and the only non-C material inside the window is separated from the non-C material outside it by a margin w. The main sheet now reads $C\,|\,B\,|\,A\,|\,C\,|\,A\,|\,B$ with interfaces at $-6, -5, -4, 0, 1$, and the auxiliary sheet is retired.
3. On the main sheet: the C band $(-4, 0)$ is flanked by A on both sides, so it dies into A; the A band $(-5, 1)$ is then flanked by B on both sides and dies into B. The word is $C\,|\,B$ with the interface at -6. Finally birth C on $(-3, 0)$ inside B and let the B band $(-6, -3)$ die into C: the interface is back at 0 and the band belongs to B.

Each move in this list has its changed region at distance at least w from every other owner, so Lemma VIII.5 supplies the information hypothesis of Lemma IX.1 or IX.2. Only three parties take part. Since a party is born once, dissolves the bands on its two sides once, and is otherwise a spectator, every party participates in a bounded number of moves.

Why a sweep rather than the dyadic hierarchy of the source? Because the final network must be placed on the chain. A link of the party network joins two parties, and every link crossing a given bond of the chain contributes a factor to that bond's dimension. In a dyadic hierarchy, every scale has a link crossing every bond (the tree of intervals has cutwidth $\log L$), so the bond dimension would be polynomial to the power $\log L$. In the sweep every link joins parties attached to blocks at most two apart, and a bounded number of links cross each bond.

Precisely. Fix $p = 12$. Let D_{24} be the constant of Lemma VIII.5 with exponent $2p = 24$ and $k_0 = 1$, and put

$$\begin{aligned} w &= \lceil D_{24} \log L \rceil, \qquad t = 20w, \\ k_* &= \max\{k \geq 0 :\ L - kt \geq 3t\}, \end{aligned} \tag{29}$$

with the blocks $B_i = [(i-1)t + 1,\ it]$ for $1 \leq i \leq k_*$ and the last block $B_{k_*+1} = [k_* t + 1, L]$, of length in $[3t, 4t)$. We assume L large enough that $t \geq 20$ and $k_* \geq 0$ (small L are handled in Section XII). For a band $[a, a + w)$ with C owning $[a - 8w, a)$ and B owning $[a + w, a + 2w)$ (and possibly more), write $x = (\cdot - a)/w$ for the scaled coordinate.

Construction X.1 (Band dissolution)**.** Let the main sheet read $C\,|\,A\,|\,B$ near a: C on $x < 0$ down to at least $x = -8$, A on $0 < x < 1$, B on $x > 1$ up to at least $x = 2$, where A, B, C are three distinct parties. Perform:

(i) Prepare an auxiliary sheet, a fresh copy of Ω held

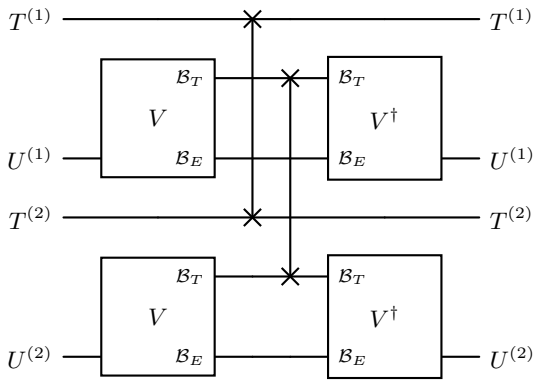


FIG. 21. The exchange correction on the two sheets: after renaming the registers inside Y, the common owner applies V to both buffers, swaps the two purifying factors, and applies the adjoints, which realizes the partial swap on T up to $4L^{-p}$.

entirely by C (a private initialization). Birth B on $(-6,-1)$; birth A on $(-5,-2)$; birth C on $(-4,-3)$.

(ii) Exchange the two sheets inside $Y = (-7,-\frac{7}{2})$ with $P_\circ = C$. Retire the auxiliary sheet (its registers stay idle at their owners).

(iii) On the main sheet: death of the C band $(-4,0)$ into A; death of the A band $(-5,1)$ into B; birth of C on $(-3,0)$ inside B; death of the B band $(-6,-3)$ into C.

If there is no left owner C (the band starts at site 1), replace the whole construction by a single death of the band into B.

Lemma X.2 (The band dissolution is a bounded change)**.** *For all sufficiently large L, each of the eight moves (seven births or deaths and one exchange) in Construction X.1 satisfies the information hypothesis of Lemma IX.1 or Lemma IX.2 with exponent $2p = 24$, so that the construction is a bounded change on the parties A, B, C with total reference-vector error at most $12L^{-12}$. Afterwards the main sheet reads $C \,|\, B$ with the interface at $x = 0$, and all other ownership is unchanged.*

Proof. We check each move; distances are in units of $w \geq D_{24} \log L$, and every changed region is an interval, so Lemma VIII.5 applies. (i) The first auxiliary birth has $E = \varnothing$ and needs no information bound. The second has changed region $(-5,-2)$ inside the B band $(-6,-1)$, and E, the non-B sites, is at distance 1. The third has changed region $(-4,-3)$ inside $(-5,-2)$, distance 1. The auxiliary word is now $C|B|A|C|A|B|C$ at $-6,-5,-4,-3,-2,-1$. (ii) The main sheet is C on $x < 0$ and the auxiliary sheet is C on $x < -6$, $(-4,-3)$ and $x > -1$. Hence $Z \cap Y = (-6,-4)$ (auxiliary B and A) and $Z \setminus Y = (-3,-1) \cup [0,\infty)$, together with $x \leq -8$ if C owns nothing beyond $x = -8$, at distance at least 1 from T. After the exchange the main sheet inside Y carries the auxiliary assignment, giving $C|B|A|C|A|B$ at $-6,-5,-4,0,1$; the auxiliary sheet inside Y is C. (iii) The C band $(-4,0)$ is surrounded by A on $(-5,-4) \cup (0,1)$, and the non-A sites after the death are $x \leq -5$ and $x \geq 1$, at distance 1. The A band $(-5,1)$ is surrounded by B on $(-6,-5) \cup (1,2)$; non-B sites after the death are $x \leq -6$ and $x \geq 2$, distance 1. The birth of C on $(-3,0)$ inside B has non-B sites at distance ≥ 2. The death of $(-6,-3)$ into C has non-C sites $x \geq 0$ at distance 3 and, if C owns nothing beyond $x = -8$, sites $x \leq -8$ at distance 2. The counts of moves and parties are as stated; each birth or death costs at most L^{-12} and the exchange at most $4L^{-12}$. □

Construction X.3 (The sweep)**.** Initially one party R_0 holds the sheet. For $k = 0, 1, \ldots, k_* - 1$, with $a = kt + 1$ the first site of block $k+1$ and the word $P_1|\cdots|P_k|R_k$:

(1) Birth R_{k+1} on $[a + t + w, L]$ inside R_k.

(2) Birth P_{k+1} on $[a + w, a + t - 1]$ inside R_k.

(3) Dissolve the band $[a, a + w - 1]$ (owner R_k) between $C = P_k$ and $B = P_{k+1}$ by Construction X.1; if $k = 0$ let the band die into P_1.

(4) Dissolve the band $[a + t, a + t + w - 1]$ (owner R_k) between $C = P_{k+1}$ and $B = R_{k+1}$ by Construction X.1.

At the end set $P_{k_*+1} = R_{k_*}$, which owns exactly the last block.

Proposition X.4 (Properties of the sweep)**.** *For all sufficiently large L, Construction X.3 is a distributed construction with the following properties.*

1. *There are $2k_* + 1 \leq 2L$ parties. The initial vector is a product by party: R_0 holds Ω, and every auxiliary copy of Ω is initialized privately at the party C of its band dissolution.*

2. *There are at most $20k_* \leq 2L$ non-private gates. Each involves at most three parties and is admissible with one shared vector or covector per monomial. Each party participates in at most 40 non-private gates over its lifetime.*

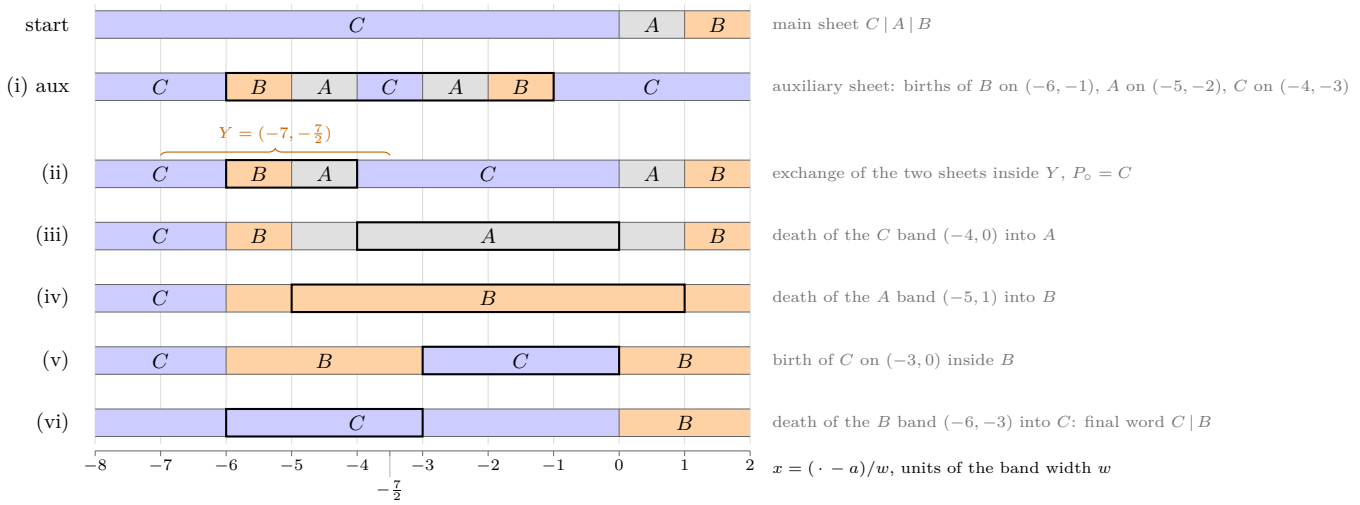


FIG. 22. The band dissolution in units of the band width: the auxiliary word after its three births (the heavier outline marks their union), the exchange window Y, and the sequence of deaths and births that leaves the band to B with the interface back at zero.

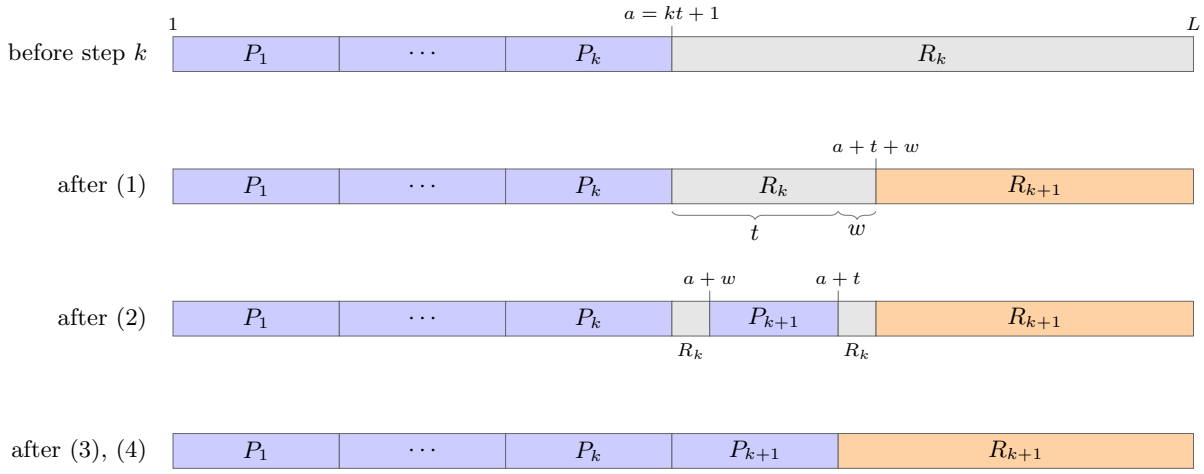


FIG. 23. One step of the sweep (for $k \geq 1$): the word before the step, after the birth of R_{k+1}, after the birth of P_{k+1}, and after the two bands of R_k have been dissolved.

3. *The final vector Ψ satisfies $\|\Psi\| \leq 1$ and $\|\Psi - \Omega \otimes \Xi\| \leq 26\, L^{-11}$, where Ξ is the tensor product of the retired auxiliary copies of Ω in their final registers and $\|\Xi\| = 1$. At the end, party P_i owns exactly the raw registers of block B_i, of dimension $q^{|B_i|} \leq q^{4t} = L^{O(1)}$, and no other party owns a raw register.*

Proof. At step k the word is $P_1|\cdots|P_k|R_k$ with R_k owning $[a, L]$, $a = kt + 1$. Move (1) has changed region $[a + t + w, L]$ and $E = [1, a - 1]$ at distance $t + w$; move (2) has changed region $[a + w, a + t - 1]$ and $E = [1, a-1] \cup [a+t+w, L]$ at distance w. Both are births by Lemma IX.1 with Lemma VIII.5. After them R_k owns the two bands $[a, a + w - 1]$ and $[a + t, a + t + w - 1]$. For (3), $C = P_k$ owns block k, of length $t = 20w \geq 8w$ to the left of a, and $B = P_{k+1}$ owns $[a + w, a + t - 1]$, of length $19w \geq 2w$, so Lemma X.2 applies (for $k = 0$, the death of $[1, w]$ into P_1 has $E = [t + 1, L]$ at distance $t - w$). For (4), $C = P_{k+1}$ owns $[a + w, a + t - 1]$ of length $19w \geq 8w$ and $B = R_{k+1}$ owns $[a + t + w, L]$ of length at least $3t - w \geq 2w$. After (4) the word is $P_1|\cdots|P_{k+1}|R_{k+1}$. The parties are $R_0, \ldots, R_{k_*}$ and $P_1, \ldots, P_{k_*}$, with $P_{k_*+1} = R_{k_*}$. The number of moves per step is at most $2 + 2 \cdot 8 = 18 \leq 20$, so at most $20k_*$ in total, and the last block has length in $[3t, 4t)$.

Participation: R_k takes part in steps $k-1$ (as the born party in (1) and as B in (4)) and k (as $P_\circ$ in (1), (2) and as A in (3), (4)); P_k takes part in step $k - 1$ (born in (2), B in (3), C in (4)) and step k (C in (3)). Each band dissolution has seven births or deaths and one exchange, which is private; so no party exceeds $2 \cdot (2 + 2 \cdot 8) = 36 \leq 40$ gates. Every birth or death is a single gate on two parties with one shared vector or covector, and every exchange consists of private gates only.

Errors: every move takes the current actual vector

within the stated error of the next reference vector, which is the tensor product of Ω on the main sheet and Ω on each auxiliary sheet in canonical coordinates (Lemmas IX.1, IX.2; retired sheets are spectators of norm one). Telescoping a product of contractions bounds the accumulated error by the sum of the per-move errors: two births and two dissolutions per step cost at most $(2 + 2 \cdot 12)L^{-12}$, so the total is at most $26k_*L^{-12} \le 26L^{-11}$. The initial vector has norm one and every gate is a contraction, so $\|\Psi\| \le 1$. □

Remark X.5 (Why no patches)**.** In the source, squares of size below $K_0 \log L$ are repainted by "small-patch rewrites" that use finite-rank cylinder projections of polynomial rank. Here the final owners hold blocks of $O(\log L)$ sites, whose register dimension $q^{O(\log L)}$ is polynomial, and the compression theorem below accepts polynomial-dimensional physical outputs. Converting a block tensor into single-site tensors is then linear algebra (Section XII). This is the second reason the adaptive patch constraint is not needed.

XI. COMPRESSION AND COLUMN EXTRACTION

In words. The sweep is a pure linear calculation ending in a partial trace over everything but the block registers. Its gates involve shared vectors of enormous dimension (purifications of half-chain marginals). The compression theorem says that if we replace, in the *density* calculation $\rho = \mathrm{Tr}_{\mathrm{private}} |\Psi\rangle\langle\Psi|$, each shared vector $|\eta\rangle\langle\tilde{\eta}|$ by an average of k independent random product operators $U_j \otimes V_j$ built from its Schmidt data and Gaussian coefficients, then for polynomial k the physical density changes by at most L^{-p} in trace norm. Expanding the error by which positions use a centered correction, each correction position involves only the two endpoint parties, and by bounded participation only boundedly many gates have to be opened for each corrected position; a Gaussian second-moment computation then shows that the expected trace norm of a term with s corrections is $k^{-s/2}$ times a fixed polynomial per correction. Nothing in this argument depends on the private dimensions or on the Schmidt ranks of the shared vectors, which is the whole point. The output is an operator tensor network with one node per party and polynomial links between parties that shared a gate. A column of it is a ket network with the same links, close to the ground vector after normalization.

The theorem and its proof are not geometric and we do not change a word of them; we restate the theorem in the form used here and sketch the estimate.

Precisely.

Theorem XI.1 (Density compression, [2, Theorem 5.2])**.** *Consider a family of distributed constructions indexed by $L \ge 2$ such that: the numbers of parties and gates are uniformly polynomial; the initial vector is a product of normalized private vectors; every gate is a contraction; each non-private gate involves at most b parties and each party participates in at most b non-private gates; each non-private gate is admissible with uniformly polynomial bounds; and the intended physical output of each party has uniformly polynomial dimension, everything else being traced out. Let ρ be the physical output density (positive, trace at most one). For every fixed $p > 0$ there is an operator tensor network σ with one node per party, a bounded number of links at each node, each link joining two parties that participate in a common non-private gate, uniformly polynomial link dimensions, and*

$$\|\sigma - \rho\|_1 \le L^{-p}.$$

The constants depend only on p, b and the uniform polynomial bounds, not on private dimensions or on the Schmidt ranks of shared vectors.

Sketch. First eliminate shared covectors: a covector $\langle\eta|$ on a pair of registers is replaced by appending $m-1$ copies of η and averaging the m cyclic permutations of the m pair registers, a contraction T_m with $\|T_m - |\eta\rangle^{\otimes m}\langle\eta|\| = m^{-1/2}$; with a common fixed garbage vector in every branch, a gate with K monomials, each with at most r covectors, is approximated in operator norm within $rm^{-1/2}\sum|c_\xi|$ by a gate with Km^r monomials using only shared vectors. Choosing m polynomial makes this error at most $\epsilon/(8M)$ per gate, M the number of non-private gates.

Next, in the density calculation each source operator $X = |\eta\rangle\langle\tilde{\eta}|$, with Schmidt data (λ_a, u_a, v_a) and $(\mu_c, \tilde{u}_c, \tilde{v}_c)$, is replaced by $\widehat{X} = \frac{1}{k}\sum_{j=1}^{k} U_j \otimes V_j$ with $U_j = \sum_{a,c}(\lambda_a\mu_c)^{1/4} g_{ac}^{(j)} |u_a\rangle\langle\tilde{u}_c|$, $V_j = \sum_{b,d}(\lambda_b\mu_d)^{1/4}\overline{g_{bd}^{(j)}}|v_b\rangle\langle\tilde{v}_d|$ and independent standard complex Gaussians $g_{ac}^{(j)}$. Then $\mathbb{E}\widehat{X} = X$, and the centered coefficients of $\Delta = \widehat{X} - X$ have diagonal covariance $k^{-1}\sqrt{\lambda_a\lambda_b\mu_c\mu_d}$ by the Gaussian pairing formula. Expanding the product over all N slot positions, $\widehat{\rho} - \rho' = \sum_{\varnothing \ne S} F_S$ where S lists the positions using a correction. For $|S| = s$, the endpoint parties number at most $2s$ and participate in at most $2bs$ gates; expand only those gates into monomials and fix the physical basis entries at those parties, at total weighted cost $Q(L)^s$ with Q a fixed polynomial. For a fixed choice, the calculation separates into an affected side, holding the corrected endpoints as free inputs, and an exterior, joined only by the exact shared vectors crossing between them, with Schmidt probability matrices $\tau, \tilde{\tau}$. The affected contractions give a block operator O of norm at most one, the corrected coefficients z_{il} have covariance $k^{-s}\sqrt{p_i\tilde{p}_l}\,\delta\delta$, and the exterior input operator is $Z = \sum_{il} z_{il}\tau^{1/2}O_{li}^{\mathsf{T}}\tilde{\tau}^{1/2}$. Schatten–Hölder with exponents $4, 2, 4$ and trace-one weights gives $\|Z\|_1 \le \|\sum_{il} z_{il}\tilde{\tau}^{1/4}O_{li}\tau^{1/4}\|_2$, whose expected square is $k^{-s}\|\widetilde{R}^{1/4}OR^{1/4}\|_2^2 \le k^{-s}$ with $R = \mathrm{diag}(p_i) \otimes \tau$, $\widetilde{R} = \mathrm{diag}(\tilde{p}_l) \otimes \tilde{\tau}$. The exterior maps and the physical matrix unit do not increase the trace norm. Hence $\mathbb{E}\|F_S\|_1 \le$

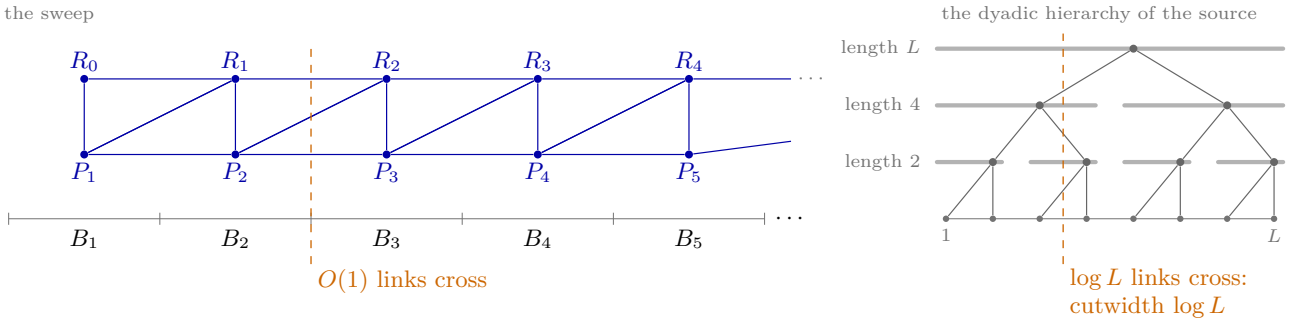


FIG. 24. Left: the party graph of the sweep placed on the chain; a bounded number of links cross any bond. Right: a dyadic hierarchy drawn on a line, where every scale contributes a link across the middle bond.

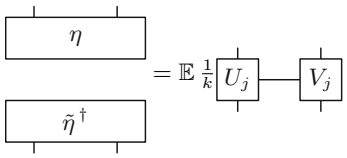


FIG. 25. The random replacement of one pair source: the operator $|\eta\rangle\langle\tilde{\eta}|$ equals the expectation of $\frac{1}{k}\sum_j U_j \otimes V_j$, the two factors joined by the sample index j.

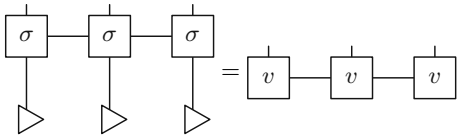


FIG. 26. Column extraction: fixing the input legs of the operator chain by a product-basis covector gives a ket chain with the same bonds.

$Q(L)^s k^{-s/2}$ and $\mathbb{E}\|\widehat{\rho}-\rho'\|_1 \le (1+Q(L)k^{-1/2})^N - 1 \le \epsilon/4$ for a polynomial k. Fix a realization. The link structure: the common monomial label of a gate is shared along a star between its participants, and each slot's sample index j is shared between its two endpoints; all private maps, private source preparations and final private traces are contracted inside each party's node. □

Lemma XI.2 (Column extraction, [2, Lemma 8.1])**.** *Let Ω be a unit vector on a finite tensor product and σ an operator with $\|\sigma - |\Omega\rangle\langle\Omega|\|_1 \le \eta < 1$. Then some product-basis column $v = \sigma|z\rangle$ is nonzero and satisfies $\min_\theta \|v/\|v\| - e^{i\theta}\Omega\| \le 2\eta$. Fixing a column in an operator tensor network leaves all virtual dimensions unchanged.*

Proof. Let $E = \sigma - |\Omega\rangle\langle\Omega|$ and $a_z = \langle\Omega|z\rangle$. Then $\sum_z \|E|z\rangle\|^2 = \|E\|_2^2 \le \eta^2$ and $\sum_z |a_z|^2 = 1$, so some z with $a_z \ne 0$ has $\|E|z\rangle\| \le \eta|a_z|$. For it, $w = v/a_z$ satisfies $\|w - \Omega\| \le \eta$, hence $w \ne 0$ and $\|w/\|w\| - \Omega\| \le 2\eta$. Fixing the input index at every node is a contraction of physical legs only. □

XII. ASSEMBLY ON THE CHAIN

In words. Apply the compression theorem to the sweep. Its parties are the block owners P_i, with physical output the $q^{|B_i|}$-dimensional block register, and the remainders R_k, with no physical output. Every link joins two parties that shared a gate, and by construction those are attached to blocks at most two apart. Put the node of R_k next to block $k+1$. Then a bond of the chain between two blocks is crossed only by links among the boundedly many parties attached to the four nearest blocks: constant congestion, and the dimension of a block-to-block bond is a bounded product of polynomial link dimensions. Finally, split each block node into one tensor per site by sequential singular value decompositions; the internal bonds are at most $q^{|B_i|}$ times the crossing dimension, still polynomial. The column-extraction lemma turns the operator network into a ket network first. Small chains are handled by the trivial exact representation.

Precisely.

Proof of Theorem I.2. Let L_0 be large enough for Proposition X.4 and for the inequalities below; for $2 \le L < L_0$ every vector on Λ is an exact matrix product state of bond dimension at most $q^{\lfloor L/2 \rfloor} \le q^{L_0/2}$, which is absorbed into C. Let $L \ge L_0$.

By Proposition X.4(3) and $\||u\rangle\langle u| - |v\rangle\langle v|\|_1 \le 2\|u-v\|$ for vectors of norm at most one, the physical output density $\rho = \mathrm{Tr}_{\mathrm{private}} |\Psi\rangle\langle\Psi|$ satisfies $\|\rho - |\Omega\rangle\langle\Omega|\|_1 \le 52L^{-11}$. The construction satisfies the hypotheses of Theorem XI.1 with $b = 40$ and physical output dimensions at most $q^{4t} \le L^{80 D_{24} \log q + 1}$, so with accuracy L^{-5} there is an operator network σ on the party graph with $\|\sigma - |\Omega\rangle\langle\Omega|\|_1 \le L^{-5} + 52L^{-11} \le 2L^{-5}$, bounded degree, and link dimensions at most $C_1 L^r$. By Lemma XI.2 some column is a nonzero ket network v with the same links and $\min_\theta \|v/\|v\| - e^{i\theta}\Omega\| \le 4L^{-5} \le L^{-1}$.

Placement. Attach P_i to block B_i and R_k to block B_{k+1}, and contract the two nodes attached to a block into one node (their mutual links cross no block bond). Two parties sharing a gate are, by the proof of Proposition X.4, among $\{P_k, P_{k+1}, R_k, R_{k+1}\}$ for some k, hence attached to the same or adjacent blocks; the bound below

allows blocks up to two apart. A bond between blocks i and $i+1$ is therefore crossed only by links incident to parties attached to blocks $i-1,\dots,i+2$; there are at most 8 such parties, each of degree at most $5\cdot 40$ (at most two star links and three sample links per three-party gate), so at most $\chi := 1600$ links cross any block bond. Route each link along the chain between its two blocks; group the indices crossing a block bond into one index of dimension at most $(C_1L^r)^\chi$. The tensor at block i is now a map from its left and right block-bond indices to the block's physical register $\mathcal{H}_{B_i}$, of dimension at most q^{4t}. Writing $B_i = \{x_1 < \cdots < x_n\}$, perform singular value decompositions successively across $x_1|x_2\cdots x_n$, $x_1x_2|x_3\cdots x_n,\dots$, treating the two block-bond indices as additional legs attached to the first and last site; the resulting single-site tensors have internal bond dimensions at most $\min\{q^j(C_1L^r)^\chi,\ q^{n-j}(C_1L^r)^\chi\} \le q^{2t}(C_1L^r)^\chi$ and the block bonds are unchanged. Thus v is a matrix product state on Λ with open boundary conditions and bond dimension at most $q^{2t}(C_1L^r)^\chi \le CL^c$, with $c = 40D_{24}\log q + r\chi$. □

Remark XII.1 (Order of constants)**.** Fix q, R, J, Δ. Theorem I.1 gives C_{int}; Lemma VIII.4 at $p = 24$ gives C_{24}, hence D_{24} and the block scale t; the number of parties, gates and the participation bound $b = 40$ are then fixed; Theorem XI.1 at accuracy L^{-5} fixes the covector stack length m, the Gaussian sample count k, and the link exponent r; the congestion χ is a numerical constant. None of these depends on the Hamiltonian or on L. The protocol uses $O(L/\log L)$ auxiliary copies of Ω, all private.

XIII. DISCUSSION

In words. *What is different from Hastings.* Hastings' area law proves that the half-chain marginal has an entanglement spectrum whose tail is exponentially small beyond rank $\exp(O(1))$, by combining a Lieb–Robinson bound, a spectral filter that approximately projects onto the ground state, and a clever bound on the approximate factorization of the filtered state. Arad–Kitaev–Landau–Vazirani replace the filter by an approximate ground-state projector built from Chebyshev polynomials with controlled Schmidt rank, and bootstrap. The present proof of Theorem I.1 has no dynamics in it at all: the gap is used only to bound the overlap of a variationally tilted or filtered vector with the ground vector, and the entropy is read off from operator norms by Hölder, a three-lines interpolation and the Araki–Lieb inequality. Its dependence on the gap is linear. The matrix product state theorem has two one-dimensional proofs here: the two-line route, which is a direct consequence of the tail bound and the area law and therefore equally free of dynamics, and the protocol route, which uses a Lieb–Robinson bound in one lemma (the reset) but in a way that has no analogue in Hastings' construction (a two-replica swap undone by a local unitary), and which otherwise consists of Uhlmann-type ownership changes, random Gaussian compression and column extraction.

What one dimension buys. Three things, each used once. A shell around an interval is crossed by $O(1)$ interactions, so the buffered discount costs a constant and needs a constant-width buffer. No three owners ever meet at a point of a line, so no encodings are needed when ownership changes. And a chain can be swept, so the party graph has bounded cutwidth. Everything else in the source is either unchanged or disappears.

What does not transfer back. The Araki–Lieb closure fails in two dimensions because the buffer is an annulus of unknown entropy; the sweep fails in two dimensions because a region being repainted has a one-dimensional boundary along which a third owner is adjacent, which the edge bands and point treatments of the source are designed to handle; and the two-line route fails because sequential truncation across nested cuts does not produce a network on the lattice. So the two-dimensional papers are not inefficient versions of this note; they solve problems that do not exist on a line.

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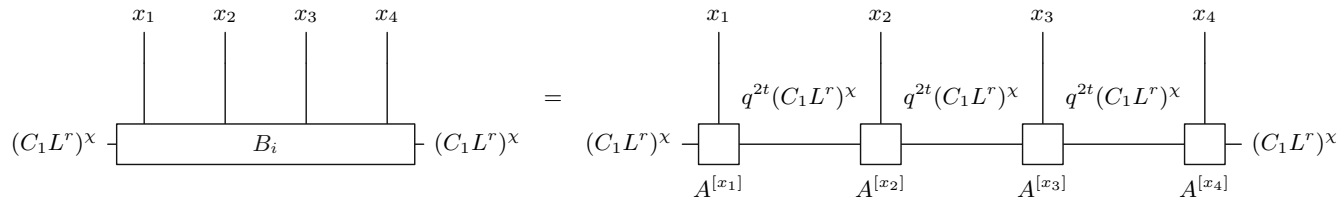


FIG. 27. A block tensor with its two block bonds and t physical legs is split into t site tensors by successive singular value decompositions; the new bonds are at most q^{2t} times the block-bond dimension.